

Financial Time Series

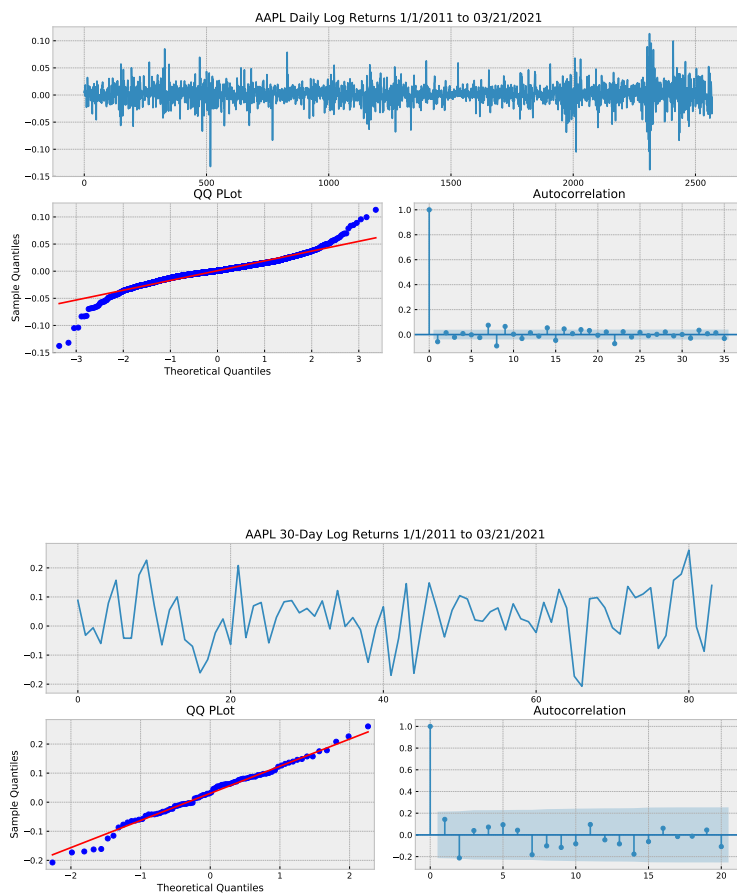
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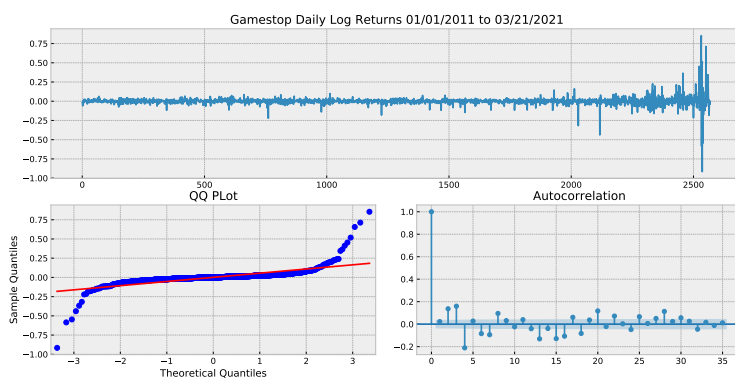
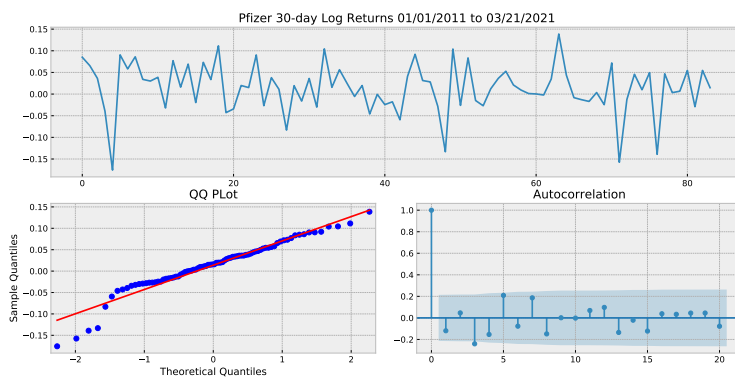
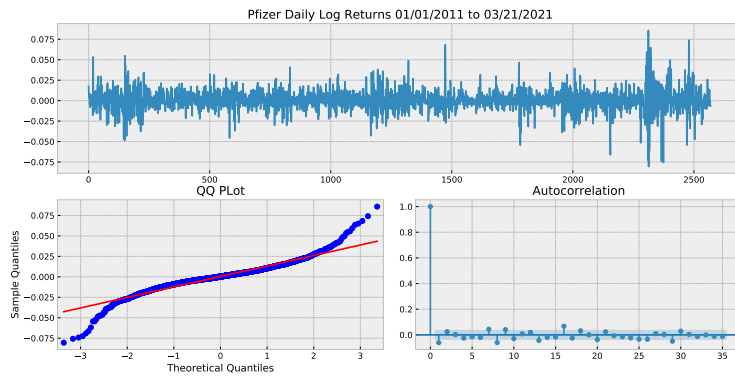
Solutions #1

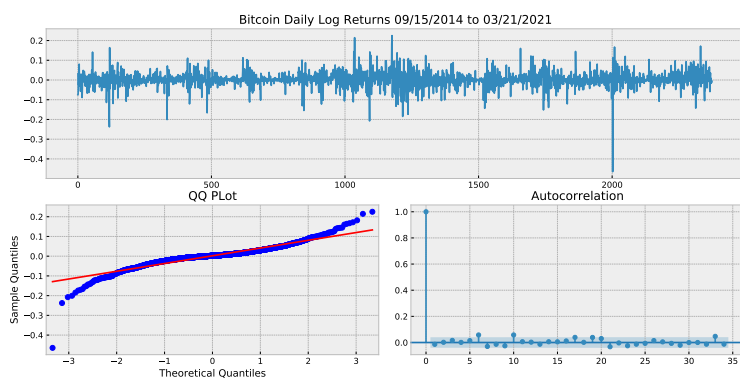
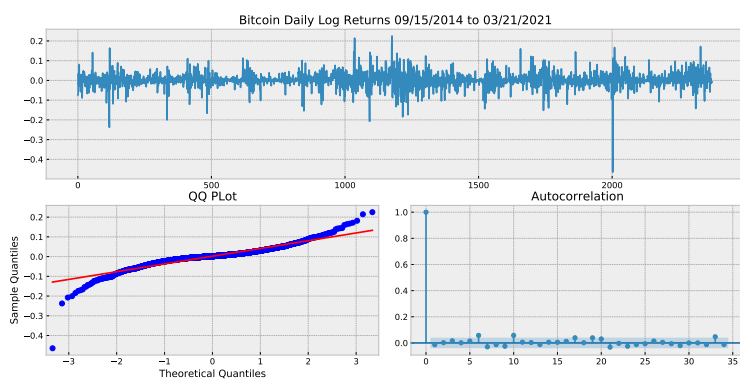
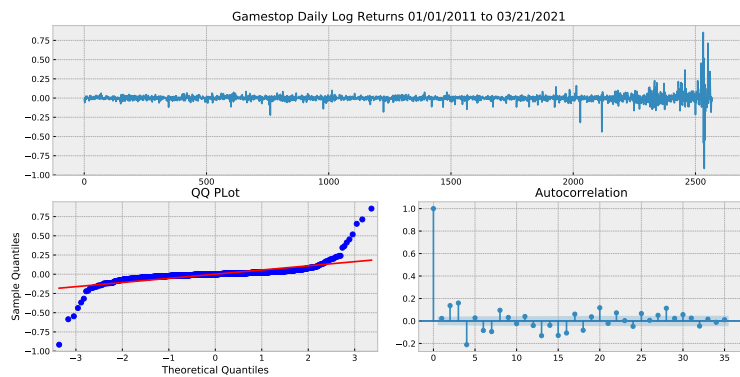
1) Practice in creating time series plots

Imitating the graphs presented in class, pick any two of the four stocks: Apple (aapl), Pfizer (pfe), Gamestop (gme) and Bitcoin (btc-usd) and create plots for the log-returns, a q-q plot for the log-returns and an acf plot for the log returns. Do your plots give evidence for or against the stylized facts of Rama Cont?

Using the code presented in lecture 1 including the function `tsplot`, we get the following plots for the four examples listed above:





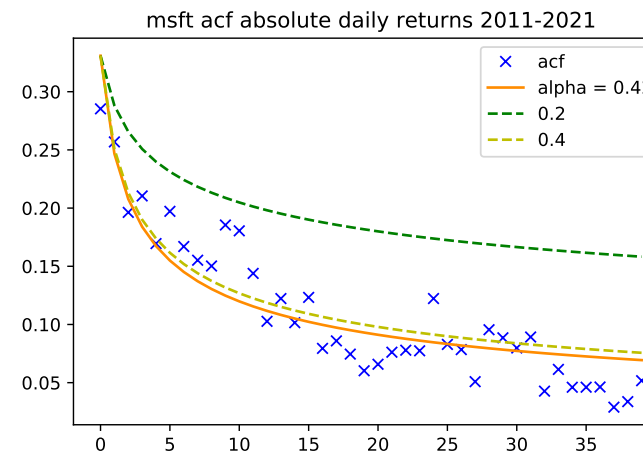
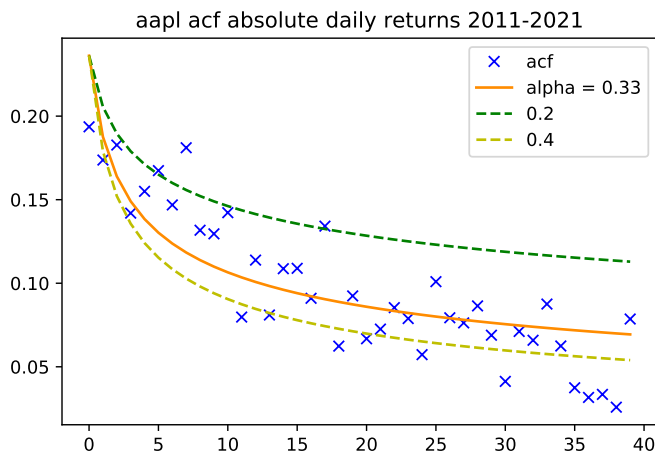
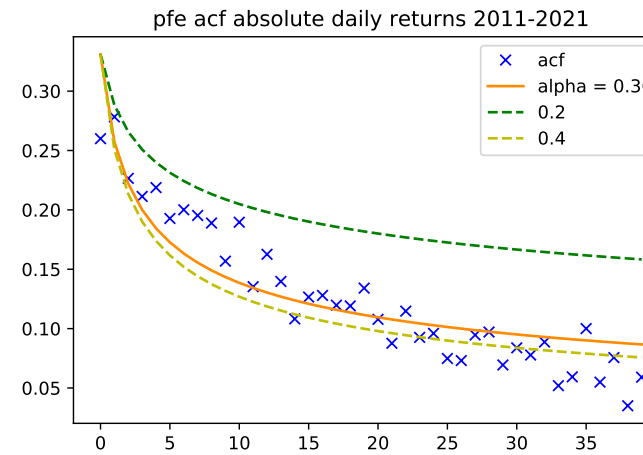
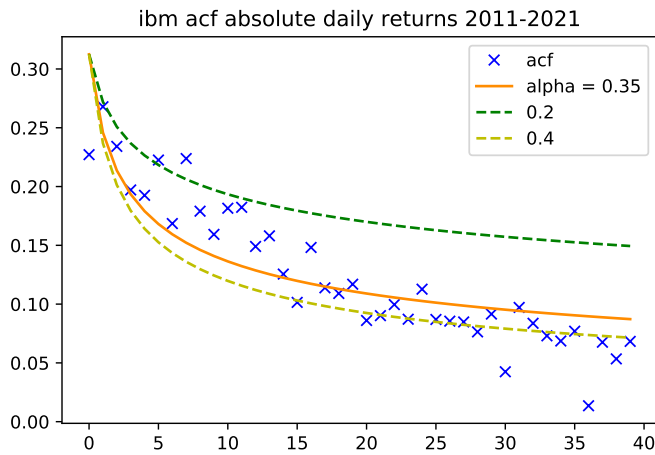


2) Behavior of absolute returns

One of the stylized principles of Rama Cont is the slow decay of absolute returns: “the autocorrelation function of absolute returns decays slowly as a function of the time lag, roughly as a power law with an exponent $\beta \in [0.2, 0.4]$.” Explore this by taking a few stocks of your choice and seeing if they follow this stylized fact.

Below are 4 figures for IBM, Pfizer, Apple and Microsoft for the absolute value of daily returns over the period 1/1/2011 to 3/21/2021. According to Rama Cont, these should decrease as a power function $f(x) = c/x * \alpha$ with exponent between $[.2,.4]$. The power function is fit using curve_fit from scipy.optimize where the fitting is done with two parameters, c and α .

We find values for α in these four cases to be IBM (.35), Pfizer (.36), Apple (.33) and Microsoft (.42), with only msft falling outside the Rama Cont region. The Rama Cont conjectured interval. The graphs below show the acf of the absolute daily returns, the fitted power function, and the power functions with $\alpha = .2$ and $.4$. The Rama Cont stylized fact seems plausible, although it is not clear that the power function is really the best description of the behavior of the acf. Still it does serve as a useful, if rough, description of this.



```

import numpy as np
import pandas as pd
import pandas_datareader as pdr
import statsmodels as smm
import statsmodels.formula.api as smf
import statsmodels.tsa.api as smt
import statsmodels.api as sm
from statsmodels.tsa.stattools import acf
import scipy
import scipy.stats as scs
from scipy.optimize import curve_fit
import matplotlib.pyplot as plt
import matplotlib as mp
import seaborn as sns

def power_fcn(x,alpha,c):
    return c/(x**alpha)

start = '2011-01-02'
end = '2021-03-21'
name = 'pfe'
stk = pdr.get_data_yahoo(name,start,end)['Adj Close']

stk_ret = np.log(stk).diff().dropna()
stk_absret = np.absolute(stk_ret)
stk_acf = acf(stk_absret)[1:]

x = np.arange(1,len(stk_acf)+1)
params = curve_fit(power_fcn,x,stk_acf)
alpha, c = params[0][0], params[0][1]
print('Alpha =',alpha)
plt.plot(stk_acf,'bx')
plt.plot(power_fcn(x,alpha,c),'darkorange')
plt.plot(power_fcn(x,0.2,c),'g-')
plt.plot(power_fcn(x,0.4,c),'y-')
plt.legend(['acf','alpha = ' + str(round(alpha,2)),'0.2','0.4'])

plt.title(name + ' acf absolute daily returns 2011-2021') plt.savefig(name + '.pdf')
plt.show()

```

- 3) Let $\{\epsilon_t, -\infty < t < \infty\}$ be a sequence of independent random variables with common mean 0 and common variance σ^2 . For each of the three processes $\{X_t, -\infty < t < \infty\}$ below, which are weakly stationary processes? For those that are, find the mean and covariance function.

- a) $X_t = a_0 + a_1\epsilon_t + a_2\epsilon_{t-1}$ for constants $a_i, i = 1, 2, 3$.
- b) $X_t = a_0 + a_1f(\epsilon_t)$ for constants a_0, a_1 and some function f .

c) $X_t = \Pi_{i=0}^k \epsilon_{t-i}$, $k \geq 1$.

a) The process is weakly stationary

i. $E(X_t) = a_0 + a_1 E(\epsilon_t) + a_2 E(\epsilon_{t-1}) = a_0 + a_1 \cdot 0 + a_2 \cdot 0 = a_0$, so the process has constant mean 0.

ii.

$$\begin{aligned} Cov(X_t, X_{t+h}) &= Cov(a_0 + a_1 \epsilon_t + a_2 \epsilon_{t-1}, a_0 + a_1 \epsilon_{t+h} + a_2 \epsilon_{t-1+h}) \\ &= a_1^2 Cov(\epsilon_t, \epsilon_{t+h}) + a_1 a_2 Cov(\epsilon_t, \epsilon_{t-1+h}) + a_1 a_2 Cov(\epsilon_{t-1}, \epsilon_{t+h}) \\ &\quad + a_2^2 Cov(\epsilon_{t-1}, \epsilon_{t+h-1}) \end{aligned}$$

Since the ϵ_t sequence is independent, all any of the covariance terms above is 0 unless the time indicies are identical, in which case it is equal to σ^2 . If $h = 0$, then the first and fourth covariances are σ^2 , the other two are 0. If $|h| = 1$, then the second term is σ^2 while the other three are 0. In all other cases, all covariance terms are 0. To summarize:

$$\gamma(h) = \begin{cases} (a_1^2 + a_2^2)\sigma^2 & h = 0. \\ a_1 a_2 \sigma^2 & |h| = 1, \\ 0 & |h| > 1. \end{cases}$$

b) Since the ϵ_t sequence is independent but not identically distributed, the X_t sequence is not, in general, stationary. If f is a linear function, then this would be a special case of (a) and, therefore, weakly stationary. But even if the first and second moments of the ϵ_t sequence are identical, it does not follow that the moment of the $f(\epsilon_t)$ sequence are identical. Moreover, there is no guarantee that the first two moments of $f(\epsilon_t)$ exist let alone be equal $\forall t$.

c) This process is weakly stationary. By independence

$$\begin{aligned} E(X_t) &= E(\Pi_{i=0}^k \epsilon_{t-i}) \\ &= \Pi_{i=0}^k E(\epsilon_{t-i}) \\ &= 0 \end{aligned}$$

$Cov(X_t, X_{t+h}) = Cov(\Pi_{i=0}^k \epsilon_{t-i}, \Pi_{i=0}^k \epsilon_{t+h-i}) = E(\Pi_{i=0}^k \epsilon_{t-i} \cdot \Pi_{i=0}^k \epsilon_{t+h-i}) - 0 \cdot 0$. If $h = 0$ then the result (again by independence) is σ^{2k} . If $h \neq 0$, then at least two of the ϵ terms in the previous expectation are to the first power. Applying independence, their expectation is 0, hence the covariance is 0. Thus the X_t process is white noise with covariance function:

$$\gamma(h) = \begin{cases} \sigma^{2k+2} & \text{if } h = 0, \\ 0 & \text{if } |h| \neq 0. \end{cases}$$

4 Let $\{\epsilon_t, -\infty < t < \infty\}$ be an independent sequence with common mean 0 and common variance σ^2 . Let

$$X_t = \epsilon_t + \epsilon_{t-1}\epsilon_{t-2}.$$

Determine if $\{X_t, -\infty < t < \infty\}$ is a white noise process and/or a MDS.

The process is weakly stationary but not an MDS. $E(X_t) = E(\epsilon_t + \epsilon_{t-1}\epsilon_{t-2}) = 0 + 0 \cdot 0$, but independence.

$$\begin{aligned} Cov(X_t, X_{t+h}) &= Cov(\epsilon_t + \epsilon_{t-1}\epsilon_{t-2}, \epsilon_{t+h} + \epsilon_{t+h-1}\epsilon_{t+h-2}) \\ &= Cov(\epsilon_t, \epsilon_{t+h}) + Cov(\epsilon_{t-1}\epsilon_{t-2}, \epsilon_{t+h}) \\ &+ Cov(\epsilon_t, \epsilon_{t+h-1}\epsilon_{t+h-2}) + Cov(\epsilon_{t-1}\epsilon_{t-2}, \epsilon_{t+h-1}\epsilon_{t+h-2}) \end{aligned}$$

Hence the autocovariance function is given by

$$\gamma(h) = \begin{cases} \sigma^2 + \sigma^4 & \text{if } h = 0, \\ 0 & \text{if } |h| \neq 0, \end{cases}$$

since the second and third terms are 0 while the first and fourth are σ^2 and σ^4 respectively. Thus the X_t process is white noise.

The process is not an MDS.

$$\begin{aligned} E(X_{t+1}|\mathcal{F}_t) &= E(\epsilon_{t+1} + \epsilon_t\epsilon_{t-1}|\mathcal{F}_t) \\ &= 0 + \epsilon_t\epsilon_{t-1}. \end{aligned}$$

Consequently, $E(X_{t+1}|\mathcal{F}_t) = \epsilon_t\epsilon_{t-1} \neq 0$, hence X_t is not an MDS.

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Solutions #2

1. Weak Stationarity

- a) (Differencing preserves weak stationarity)

Suppose $\{X_t, -\infty < t < \infty\}$ is a weakly stationary process. Define a new process $Y_t = (1 - B)X_t = X_t - X_{t-1}$. Show that $\{Y_t, -\infty < t < \infty\}$ is also weakly stationary and find the mean, variance and autocovariance function of the Y_t process.

- b) (Linear combinations of independent weakly stationary processes are weakly stationary)

Suppose $\{X_t^{(i)}, -\infty < t < \infty, i = 1, 2\}$ are weakly stationary independent. Show that $\{Y_t, -\infty < t < \infty\}$ where $Y_t = aX_t^{(1)} + bX_t^{(2)} + c$ is also weakly stationary.

Solution

- a) Suppose X_t process is weakly stationary with autocovariance function $\gamma_X(h)$ and $Y_t = X_t - X_{t-1}$. Then

$$\begin{aligned} \text{Cov}(Y_t, Y_{t+h}) &= \text{Cov}(X_t - X_{t-1}, X_{t+h} - X_{t-1+h}) \\ &= \text{Cov}(X_t, X_{t+h}) - \text{Cov}(X_t, X_{t-1+h}) - \text{Cov}(X_{t-1}, X_{t+h}) + \text{Cov}(X_{t-1}, X_{t-1+h}) \\ &= \gamma_X(h) - \gamma_X(h-1) - \gamma_X(h+1) + \gamma_X(h) \end{aligned}$$

Thus $\gamma_Y(h) = 2\gamma_X(0) - \gamma_X(h-1) - \gamma_X(h+1)$ which is independent of t . Also the mean $E(Y_t) = 0$, so the mean and variance are constant and the covariance function depends only upon the time lag. Hence Y_t is weakly stationary.

- b) The calculation is similar to part a). Because $\{X_i, i = 1, 2\}$ are weakly stationary, they have constant mean μ_i , constant variance σ_i^2 and are independent. Consequently, $E(Y_t) = a\mu_1 + b\mu_2 + c$ and $\text{Var}(Y_t) = a^2\sigma_1^2 + b^2\sigma_2^2$, both independent of t . Since the constant mean of Y_t and the constant c will not impact the autocovariance function of the Y process, we compute

$$\begin{aligned} \text{Cov}(Y_t, Y_{t+h}) &= \text{Cov}(aX_t^{(1)} + bX_t^{(2)}, aX_{t+h}^{(1)} + bX_{t+h}^{(2)}) \\ &= a^2\text{Cov}(X_t^{(1)}, X_{t+h}^{(1)}) + b^2\text{Cov}(X_t^{(2)}, X_{t+h}^{(2)}) + 2ab(0) \\ &= a^2\gamma_{X^{(1)}}(h) + b^2\gamma_{X^{(2)}}(h). \end{aligned}$$

Thus the autocorrelation function of $\{Y_t\}$ is independent of t . Consequently, the process is weakly stationary.

2. Introduction to the Yule Walker Equations

In lecture #3, we found that for an AR(2) process, $\{X_t, -\infty < t < \infty\}$ with $\{\epsilon_t, -\infty < t < \infty\}$ a $\text{WN}(0, \sigma_\epsilon)$ process, there was linear relationship between the model coefficients ϕ_1, ϕ_2 and the autocovariance function $\gamma(h)$ recalling that $\gamma(0) = \sigma_\epsilon^2$ and $\gamma(-h) = \gamma(h)$. This was used to find an iteration by which one could express $\gamma(h)$ or $\rho(h)$ as a function of ϕ_1 and ϕ_2 for any h .

- a) Turning this around, find two linear equations for ϕ_1, ϕ_2 in terms of ρ_1 and ρ_2 . Express those two equations in matrix form. Solve to find a matrix expression for ϕ_1 and ϕ_2 .
- b) Suppose we have estimates of $\gamma(1)$ and $\gamma(2)$, namely $\hat{\gamma}(1)$ and $\hat{\gamma}(2)$. One could plug them into your expression from part a) to obtain estimates for ϕ_1 and ϕ_2 . Simulate 1,000 data points from an AR(2) sequence with iid $N(0, \sigma^2)$ errors and using those data, estimate ϕ_1 and ϕ_2 . Use

$\phi_0 = 0, \phi_1 = .33, \phi_2 = .33, \sigma_\epsilon^2 = .5$. Note, you can use the Python routine `statsmodels.regression.linear_models.yule_walker(data, order=2, method='mle')` as this 2x2 system is a very simple version of the general Yule-Walker equations to be covered in class.

- c) The accuracy of this method (assuming the data actually conforms to the model) depends upon σ_ϵ^2 and n , the size of the data. Repeat part b) with $n = 100$ and $n = 500$ to get an idea of how the error of estimation changes with the sample size.

Solution

- a) It is convenient to remove any mean term as it will not change the autocorrelation of the process, so we set $Y_t = X_t - \mu$, where $E(X_t) = \mu$. $\rho_X(i) = \rho_Y(i)$. As was presented in lecture, we start with the AR(2) model,

$$y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t.$$

Multiplying across by first Y_{t-1} then Y_{t-2} , taking expectations and dividing by $\gamma(0)$, we find

$$\rho(1) = \phi_1 \rho(0) + \phi_2 \rho(1) + 0,$$

$$\rho(2) = \phi_1 \rho(1) + \phi_2 \rho(0) + 0,$$

since Y_s and ϵ_t are uncorrelated for $s < t$. Recalling that $\rho(0) = 1$, these two equations can be put into matrix form with $\boldsymbol{\rho} = (\rho_1, \rho_2)^T$, $\boldsymbol{\phi} = (\phi_1, \phi_2)^T$, and

$$\mathbf{R} = \begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix},$$

$$\boldsymbol{\rho} = \mathbf{R}\boldsymbol{\phi}$$

This can be solved by

$$\boldsymbol{\phi} = \mathbf{R}^{-1}\boldsymbol{\rho}.$$

- b)c) In a single run, the results were $\hat{\phi}_1 = .3036$ and $\hat{\phi}_2 = .3391$. One would need to run this routine m times to get some idea of the variability of the estimate. Below in a table are the estimates for $m = 10$ runs using $\sigma = .5$ for $n = 100, 500, 1000$. The results are the mean over 10 runs for each of the 3 cases. The estimates are given along with the estimated standard deviations (the sample standard deviation, not the standard error of the parameter estimates, which would need to be divided by $\sqrt{n}$).

n	100	500	1000
$\hat{\phi}_1$	.3544 (.1092)	.3518 (.0324)	.3370 (.0305)
$\hat{\phi}_2$	.2840 (.0849)	.3150 (.0434)	.3055 (.0280)

3. MA(1)

Consider a MA(1) process with mean 0,

$$X_t = \theta_1 \epsilon_{t-1} + \epsilon_t,$$

for ϵ_t a $WN(0, \sigma^2)$. Pick your own coefficient θ_1 and let $n = 1000$. We worked out the autocovariance and autocorrelation function for some specific cases including MA(1). If one has a moment estimator for $\gamma(1)$, then one can use this to find a moment estimator for θ_1 . Note, this can be done for higher-order MA processes, but the resulting equations are non-linear and difficult to solve. Simulate the MA(1) series of (length 1,000) 10 times with $\sigma^2 = 1$ each time using the simulated data to estimate θ . Report your 10 estimates and the variability of those estimates.

Solution

The MA(1) process is given by the equation

$$X_t = \mu + \theta_1 \epsilon_{t-1} + \epsilon_t.$$

As before, we set $\mu = 0$ since it will not change the autocovariance function. We multiply both sides by X_{t-1} and take expectations. This gives $E(X_t X_{t-1}) = \theta_1 E(X_{t-1} \epsilon_{t-1}) + E(\epsilon_t X_{t-1})$. $E(\epsilon_t X_{t-1}) = 0$, because X_{t-1} is a linear function of the ϵ_s , $s < t$, so X_{t-1} and ϵ_t are uncorrelated. Thus $\gamma(1) = \theta_1 \sigma_\epsilon^2$. Dividing by $\gamma(0) = (1 + \theta_1^2) \sigma_\epsilon^2$ we find

$$\rho(1) = \frac{\theta_1}{1 + \theta_1^2}$$

. This gives a quadratic equation for θ_1 as a function of $\rho(1)$. If we have a moment estimate of $\rho(1)$, then we can plug it into the solution of the quadratic to get an estimate of θ_1 . Specifically,

$$\hat{\theta}_1 = \frac{1 - \sqrt{1 - 4r^2}}{2r},$$

where r is the moment estimate of $\rho(1)$. We pick the root so that $|\theta_1| < 1$, so the process is invertible.

The results from 10 simulations of θ_1 using this method with $\theta_1 = .25$ and $\sigma_\epsilon = 1$ had a mean value of .2660 and a standard deviation of .0249 with a range of [.2395,.3109].

4. Spurious Regression

The problem of spurious regression (and spurious correlation) arises especially in economics where one performs a regression analysis with non-stationary time series data, and most economic time series are non-stationary. The simplest situation is where one has two non-stationary time series that are independent and regresses one on the other. Because the two series, while being independent, are both related to time, a significant but spurious correlation or regression can be found. Time is a confounding factor. The first report of this phenomenon dates back to G. Udny Yule in 1926, “Why do we Sometimes get Nonsense-Correlation between Time Series? – A Study in Sampling and the Nature of Time-Series.” The most cited paper on this topic is the 1974 paper by Granger and Newbold, “Spurious Regressions in Econometrics,” (both papers are in the Course Papers folder). The simplest version of this phenomenon is where $\{X_t\}$ and $\{Y_t\}$ are independent random walks (which are non-stationary). That is $X_t = \sum_{i=1}^t Z_i^{(1)}$ and $Y_t = \sum_{i=1}^t Z_i^{(2)}$ where $\{(Z_i^{(1)}, Z_i^{(2)})\}$ are iid with independent $N(0, \sigma_j^2)$, $j = 1, 2$ distributions. Consider the model

$$Y_i = \beta_0 + \beta_1 X_i, \quad 1 \leq i \leq t.$$

Clearly because the X and Y sequences are independent, $\beta_1 = 0$; however, one can still test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$. Granger and Newbold did this 100 times with simulated data and found that roughly 75% of the trials resulted in rejection of H_0 . In this problem let $\sigma_i^2 = 1$, $i = 1, 2$, simulate $\{(X_t, Y_t), 1 \leq t \leq 50\}$ and test whether H_0 is significant at the 5% level. Do this 100 times and report the results.

Solution

Granger and Newbold fit a simple linear regression model $Y = \beta_0 + \beta_1 X + \epsilon$ with a sample size of 50 for two independent random walks for X and Y . They tested $H_0 : \beta_1 = 0$ at the .05 α level. They reported that “three-quarters” of the 100 trials resulted in rejection, indicating a spurious regression. While the problem called for replication of the results, the results below are for a larger sample size ($n = 10,000$), show both $\alpha = .05$ and .01 and use the size of the data to be 50, 100 and 250. One can see that the “three-quarters” probability of a spurious result cited by Granger and Newbold is too high (presumably due to sampling variability) as “two-thirds” is closer. Also increasing the sample size leads to more frequent spurious regression. Also reducing the α level from .05 to .01 reduces the frequency of spurious regression; however, it is still a very frequent phenomenon. The python program below will produce the results in the table.

α	$n = 50$	$n = 100$	$n = 250$
.05	.667 (.005)	.761 (.004)	.847 (.004)
.01	.574 (.005)	.689 (.005)	.803 (.004)

```

import numpy as np
import statsmodels.api as sm

n=10000
reject = 0
for i in range(n):
    z1 = np.random.randn(50)
    z2 = np.random.randn(50)
    x = np.cumsum(z1)
    y = np.cumsum(z2)
    x_one = sm.add_constant(x)
    model = sm.OLS(y,x_one).fit()
    if model.pvalues[1] < 0.05:
        reject = reject+1

print(rej)
p = rej/n
print(np.sqrt(p*(1-p)/n))

```

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Solutions #3

1. Download data on three-month treasury bill rates from the St. Louis Federal Reserve for 1/1/85 to 12/31/2005. You can download the series from:
research.stlouisfed.org/fred2/series/TB3MS?cid=116
 - a) Plot the data and indicate whether the data series seems to be stationary.
 - b) If the series is not apparently stationary, difference the series as few times as needed to reject the presence of a unit root using the Dickey Fuller test.
 - c) Fit an ARIMA(p,d,q) model to the series using the order_selection command, keeping d at the value you chose in the previous part. If the original series was considered to be stationary, then set $d = 0$.
 - d) Run diagnostic tests on the model. Does it seem to be a reasonable fit?

SOLUTION

- a) The following Python script will give answers to the questions below. Clearly the series is not stationary.

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.stattools import arma_order_select_ic
from statsmodels.tsa.arima_model import ARIMA

us_tb = pd.read_csv('TB3MS.csv')
us_tb = us_tb.iloc[(1985-1934)*12:-(2019-2005)*12-3]

adfuller(us_tb['TB3MS'].values)

adfuller(np.diff(us_tb['TB3MS'].values, 1))
plot_acf(np.diff(us_tb['TB3MS'].values, 1))
plt.show()
plot_pacf(np.diff(us_tb['TB3MS'].values, 1))
plt.show()
arma_order_select_ic(np.diff(us_tb['TB3MS'].values, 1), ic=['aic', 'bic'])
model = ARMA(np.diff(us_tb['TB3MS'].values, (3, 0)).fit(dis=0))
model = ARIMA(us_tb['TB3MS'].values, (1, 1, 0)).fit(dis=0)
```

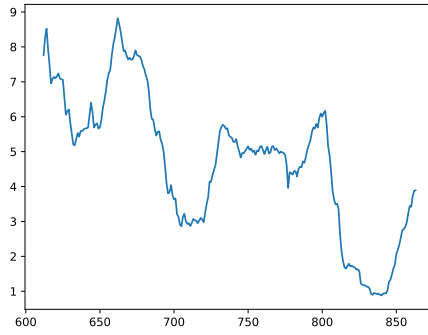
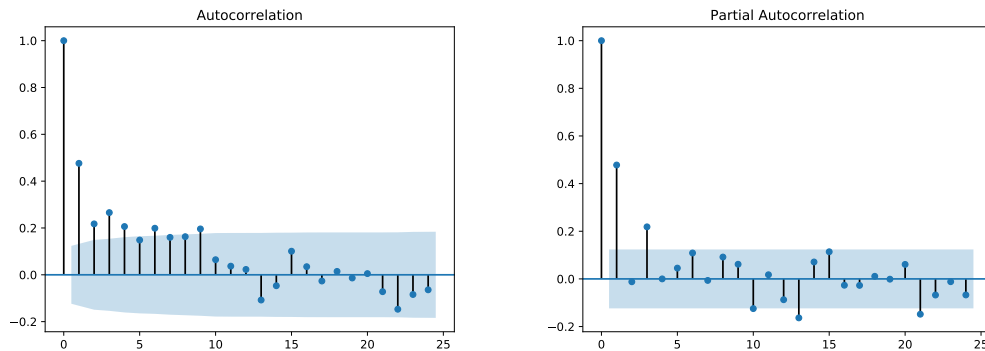


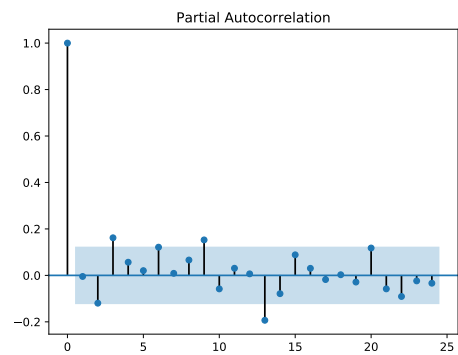
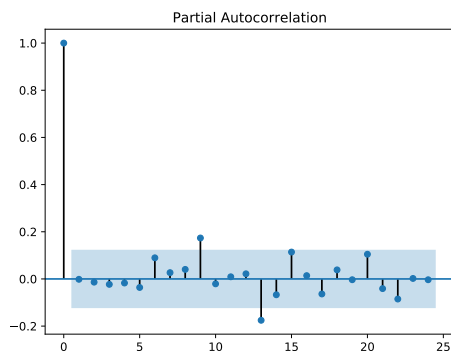
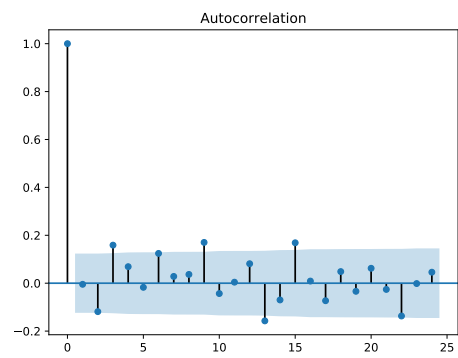
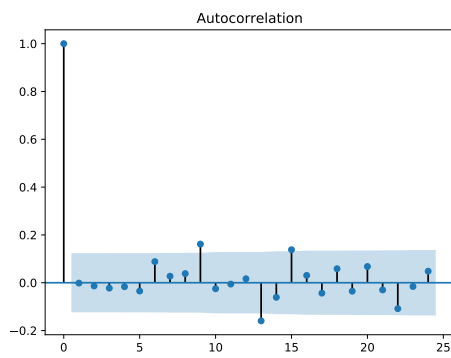
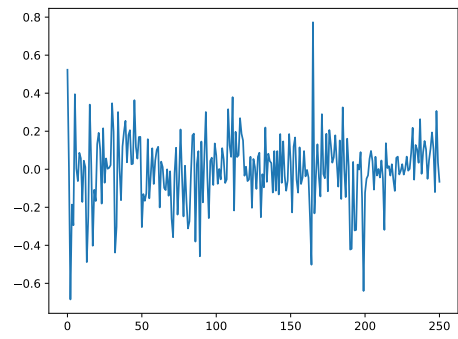
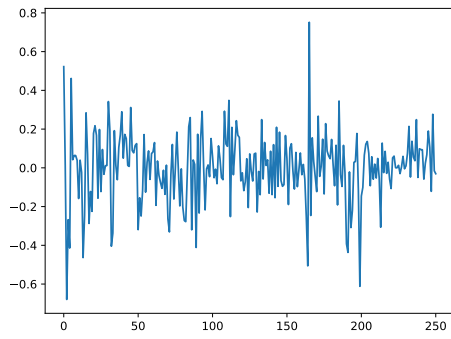
Figure 1: Raw data

- b) The Dickey Fuller test on the raw data does not reject the null hypothesis of a unit root as the p-value is .146: $(-2.3835424188318792, 0.1464354168524229, 10, 241)$. When the data is differenced one time, we get a p-value of $1.04e-06$, and reject the null hypothesis. $(-5.639612366776882, 1.0437937963045998e-06, 2, 248)$. We now conduct analysis with the differenced data.
- c) We next show the ACF and PACF for the differenced sequence. The PACF shows that we need to include at least 3 lags for an AR model.



The ARMA order select command gives is (3,0) for AIC and (1,0) for BIC, which means ARIMA(3,1,0) or ARIMA(1,1,0) for the original data set. The BIC result is suspicious since the PACF above indicated we need more lags than 1.

- d) We show the residuals, the ACF and the PACF figures side-by-side for the ARIMA(3,1,0) on the left and ARIMA(1,1,0) on the right. From the ACF and PACF, it is clear that the ARIMA(3,1,0) model is the best.

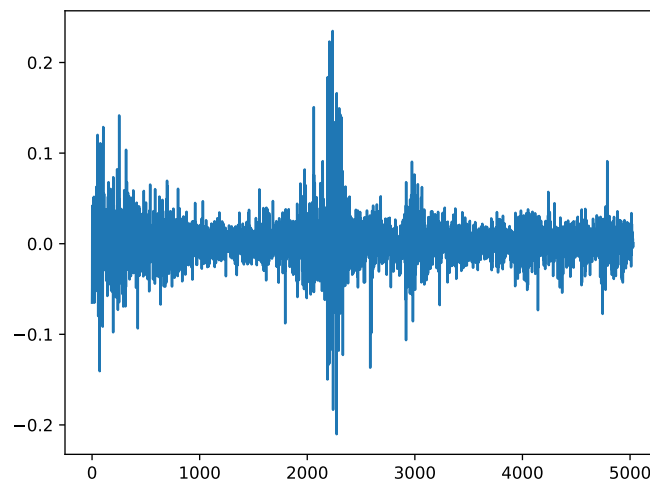


2. Consider any stock of your own choosing and any time period of at least 10 years length.
 - a) Plot the one-day log returns using the Adjusted Close data.
 - b) Using the adjusted Dickey Fuller test, assess whether the series is weakly stationary. If not, can you make it stationary by differencing?
 - c) Plot the ACF and PACF. What do you see?
 - d) Use the Arma order_select command in Statsmodels. Using the aic and bic criteria, find the suggested appropriate ARMA model(s),
 - e) From the models selected and other similar models, present the results from the best model in your judgment. Does it provide a good fit to these data? Does it pass the Ljung-Box and the adjusted Dickey Fuller tests?

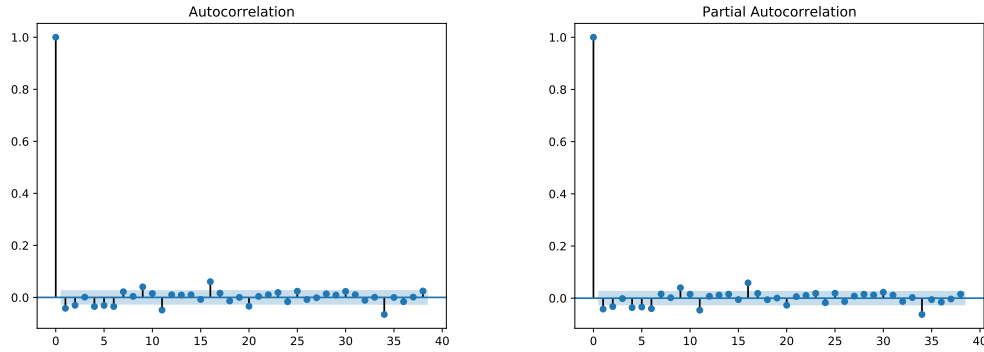
Solution

Using Goldman-Sachs for 1/1/2000 to 12/31/2019.

- a) Plot the one-day log returns using the Adjusted Close data.



- b) Using the adjusted Dickey Fuller test, assess whether the series is weakly stationary. If not, can you make it stationary by differencing?
 The adjusted Dickey-Fuller test clearly rejects the null hypothesis of a unit root with tiny p-value of 1.26e-29.
 (-16.782971758161047, 1.2598274349939462e-29, 15, 5014,)
- c) Plot the ACF and PACF. What do you see?



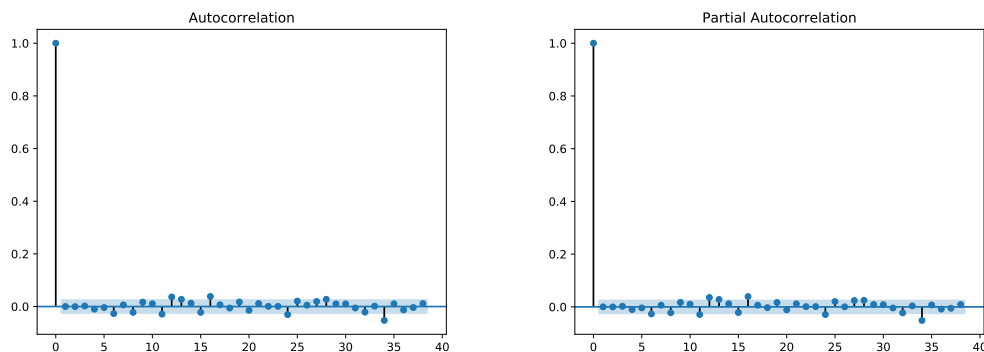
- d) Use the `Arma_order_select` command in `Statsmodels` to find the suggested appropriate ARMA model(s),

```
res = sm.tsa.arma_order_select_ic(logret_ibm_shf, ic=['aic', 'bic'], trend='nc')
print(res.aic_min_order) (4,2)
print(res.bic_min_order) (1,1)
```

- e) From the models selected, present the results from the best model. Does it provide a good fit to these data?

The two selected models based on the AIC and BIC provide very good fits to the data, at least from looking at the ACF and PACF plots of the residuals for both. All of the coefficients are essentially non-significant (I didn't print out the actual confidence interval endpoints), and are in most cases tiny. We could pick either, and visually (4,2) is a bit better than (1,1); however, at the cost of 6 parameters versus 2. It would seem that an ARMA(1,1) model provides a very good fit to the data, and it has been selected by the BIC criterion. Nevertheless, there are clear signs of stochastic volatility, so we could revisit this example when we have learned about GARCH models.

(4,2) model



ACF

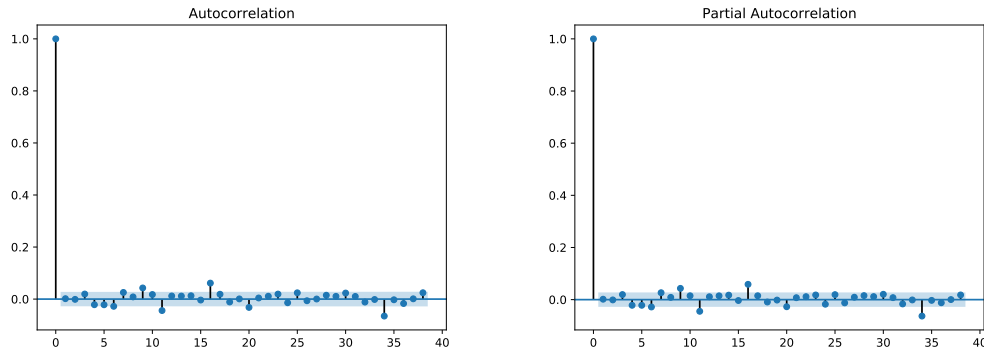
```
(array([ 1.00000000e+00,  1.79461280e-04, -2.44975335e-04,  2.17206131e-03, -1.00356145e-02,
        -3.85632875e-03, -2.66587744e-02,  6.33609644e-03, -2.20378382e-02,  1.68499359e-02,
        1.06757752e-02]), array([[ 1.00000000e+00,  1.00000000e+00]]),
```

PACF

```
(array([ 1.00000000e+00,  1.79496965e-04, -2.45105006e-04,  2.17344574e-03, -1.00444915e-02,
        -3.85567065e-03, -2.67021028e-02,  6.39801883e-03, -2.21982616e-02,  1.69631657e-02,
```


1.00800419e-02]], array([[1.00000000e+00, 1.00000000e+00],

(1,1) model



ACF

(array([1. , 0.00159288, -0.00102386, 0.01947379, -0.02160683, -0.02179755, -0.02776337, 0.02558894, 0.00840609, 0.04280169, 0.01761185]), array([[1.00000000e+00, 1.00000000e+00],

PACF

(array([1. , 0.00159319, -0.00102681, 0.01948875, -0.02169594, -0.0217181 , -0.02818144, 0.02655826, 0.00870158, 0.04319192, 0.01487181]), array([[1.00000000e+00, 1.00000000e+00],

3. (Ruppert and Matteson, problem 16, page 359)

Suppose you fit an AR(2) model to a time series Y_t , $1 \leq t \leq n$ and the estimates were $\hat{\mu} = 100.1$, $\hat{\phi}_1 = 0.5$ and $\hat{\phi}_2 = 0.1$. The last three observations were $Y_{n-2} = 101.0$, $Y_{n-1} = 99.5$, and $Y_n = 102.3$. What are the forecasts of Y_{n+1} , Y_{n+2} and Y_{n+3} ?

Solution

Using the AR(2) model $Y_n = \mu + \phi_1(Y_{n-1} - \mu) + \phi_2(Y_{n-2} - \mu) + \epsilon_n$, plugging in the estimates for μ , ϕ_1 , ϕ_2 and estimating the future ϵ terms to be 0, we find

$$\hat{Y}_{n+1} = .4 \cdot \mu + .5Y_n + .1Y_{n-1} = 40.04 + .5 \cdot 102.3 + .1 \cdot 102.3 = 101.14$$

$$\hat{Y}_{n+2} = .4 \cdot \mu + .5\hat{Y}_{n+1} + .1Y_n = 40.04 + .5 \cdot 101.14 + .1 \cdot 102.3 = 100.84$$

$$\hat{Y}_{n+3} = .4 \cdot \mu + .5Y_{n+2} + .1Y_{n+1} = 40.04 + .5 \cdot 100.84 + .1 \cdot 101.14 = 100.574$$

4. (Ruppert and Matteson, problem 12, page 357)

The MA(2) model $Y_t = \mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}$ was fit to data and the estimates are $\hat{\mu} = 45$, $\hat{\theta}_1 = 0.3$, and $\hat{\theta}_2 = -0.15$. Suppose the last two values of the observed time series and residuals are $Y_{n-1} = 39.8$, $Y_n = 42.7$, $\hat{\epsilon}_{n-1} = -4.3$ and $\hat{\epsilon}_n = 1.5$. Find the forecasts of Y_{n+1} and Y_{n+2} .

Solution

$$Y_t = \mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}.$$

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\epsilon}_{t+1} + \hat{\theta}_1\epsilon_t + \hat{\theta}_2\epsilon_{t-1} = 45 + 0 + .3(1.5) + (-.15)(-4.3) = 46.095$$

$$\hat{Y}_{t+2} = \hat{\mu} + \hat{\epsilon}_{t+2} + \hat{\theta}_1\hat{\epsilon}_{t+1} + \hat{\theta}_2\epsilon_t = 45 + 0 + 0 + (-.15)(1.5) = 44.775$$

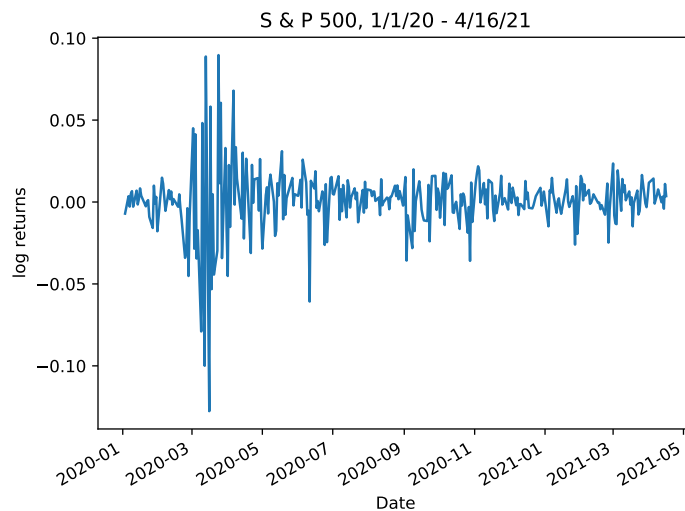
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Solutions #4

1. (Forecasting stock prices)

Download S&P 500 index data for 1/1/2020 to 4/16/2021. Using log-returns check that the time series is weakly stationary. Using data for 1/1/2020 to 4/1/2021 fit an ARIMA model and forecast the index from 4/2/2021 to 4/16/2021.



```
smt.stattools.adfuller(logret)
(-4.727019298706814, 7.484883471668374e-05, 8, 315, '1%'
%-3.451281394993741 '5%': -2.8707595072926293, '10%': -2.571682118921643, 1224.9259274784883)
```

The Dickey Fuller statistics gives a p-value of 7.485 e-05 which is far below 1%, hence we clearly reject the unit root and conclude that the series is weakly stationary.

Using the entire data set and the command

```
smt.tsa.stattools.arma_order_select_ic(logret,ic=('aic','bic'))
```

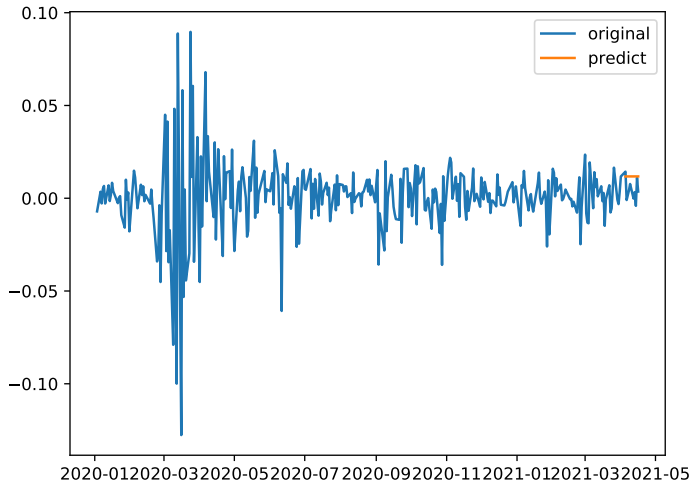
we find the best fit using either aic or bic to be ARMA(2,2).

To predict, we divide into a training and testing set and use pmdarima to fit the best model. This turns out to be ARIMA(0,0,2)

```
train = logret[:'2021-4-1']
test = logret['2020-4-1':]
model=pm.auto_arma(train, seasonal=False, trace = True)
Best model: ARIMA(0,0,2)(0,0,0)[0]
forecast = model.predict(test.shape[0])
```

The figure below gives a graphic of the S&P 500 and the prediction, done for log-returns. By computing `S.and_P[0]*ac(np.exp(np.cumsum(logret)))`, this will map the log-returns and the prediction to their natural values.

This result is quite disappointing. The problem is that the “best” model is ARIMA(0,0,2), a MA(2) model. Recall that when forecasting with a MA model, once the q parameter is reached (2 in this case), the predictions will revert to the mean. Recognizing this, one should search for a model that gives a reasonable fit and has more predictive power.



2. (Simulating ARCH(1))

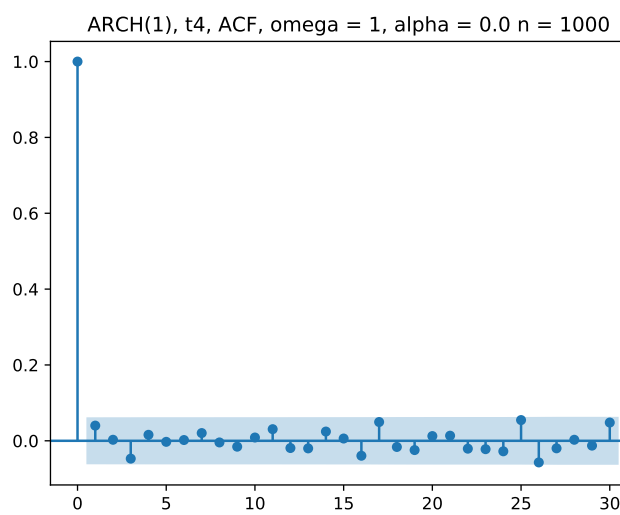
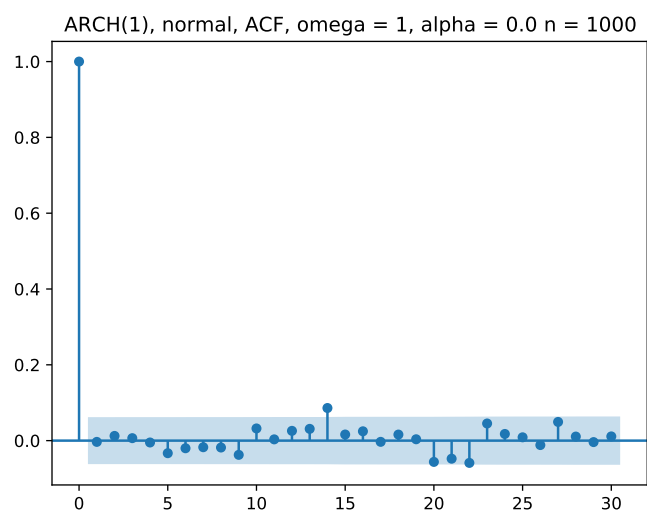
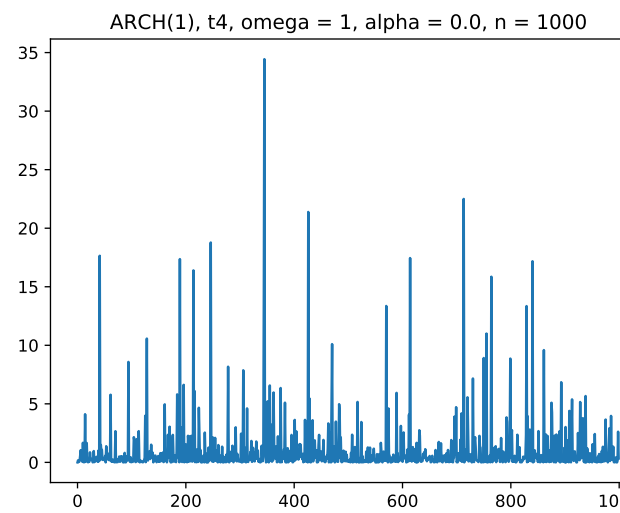
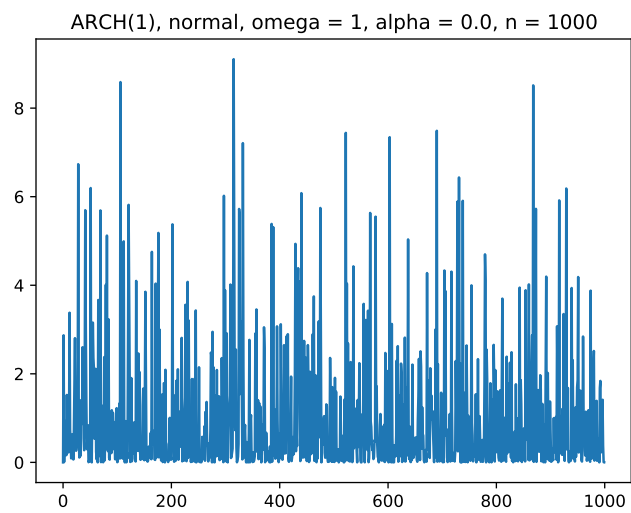
Consider an ARCH(1) model

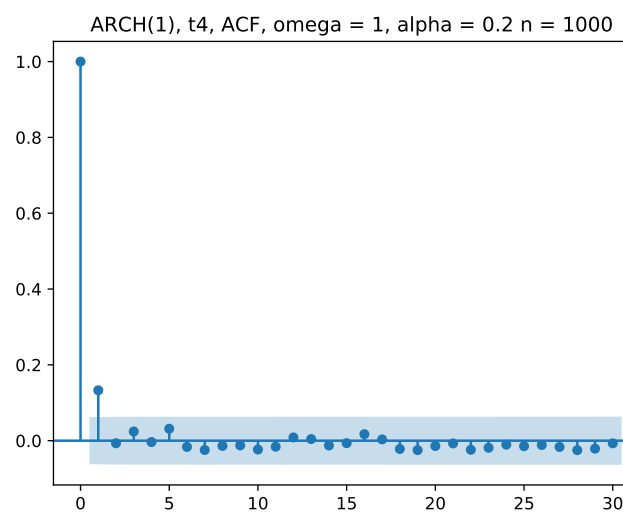
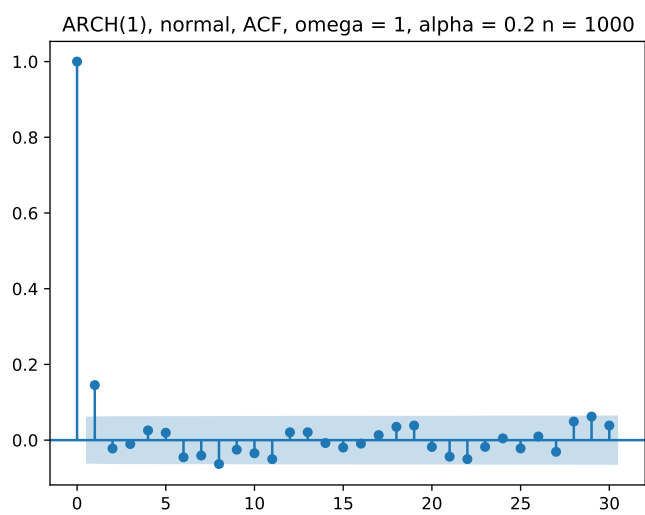
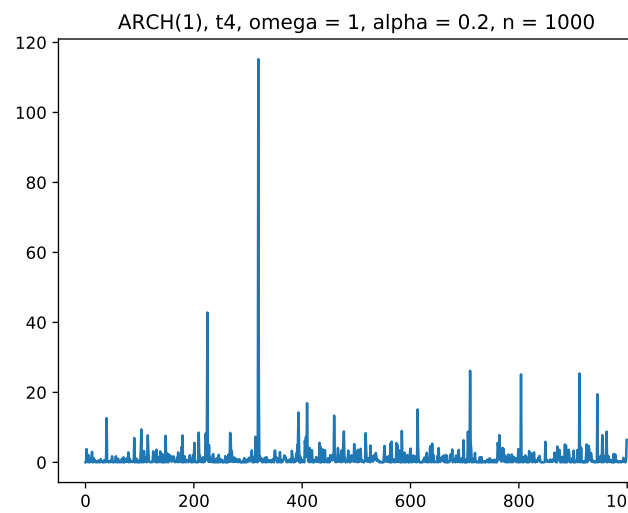
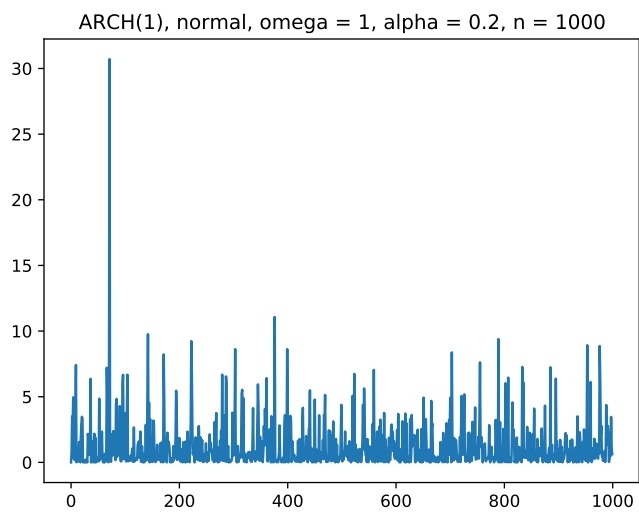
$$a_t = \epsilon_t \sigma_t,$$

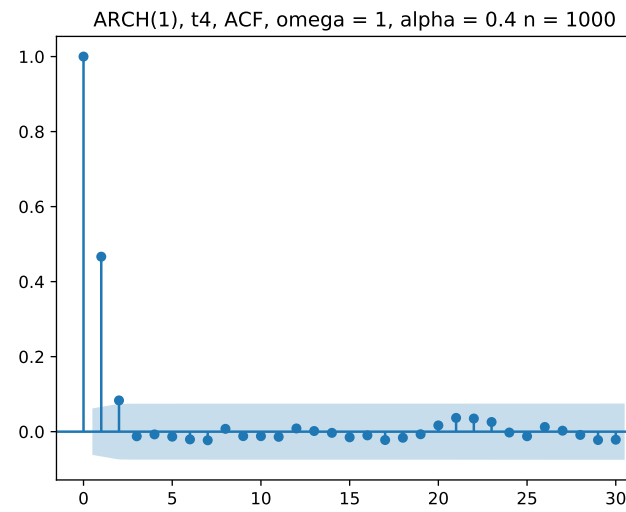
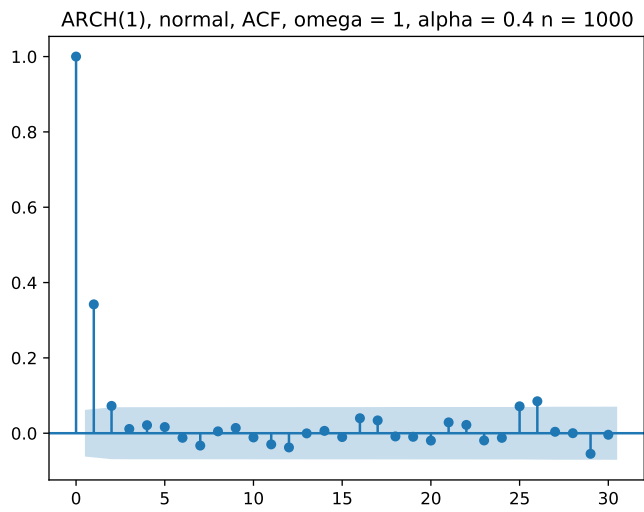
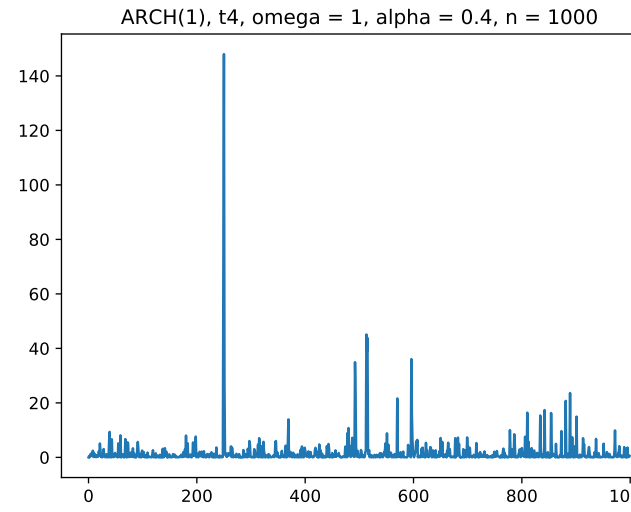
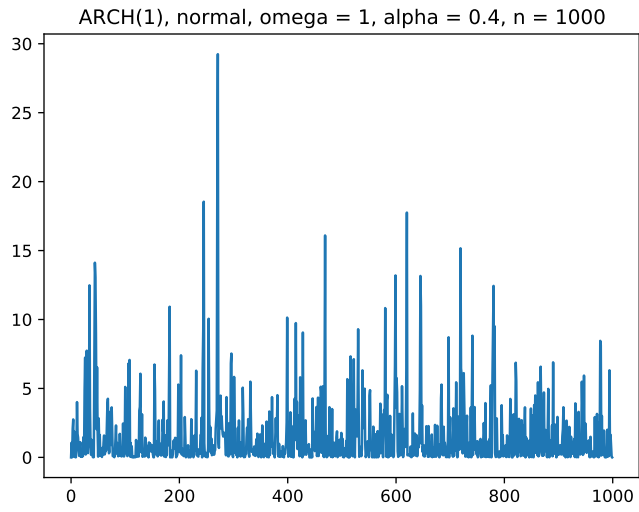
$$\sigma_t^2 = 1 + \alpha a_{t-1}^2,$$

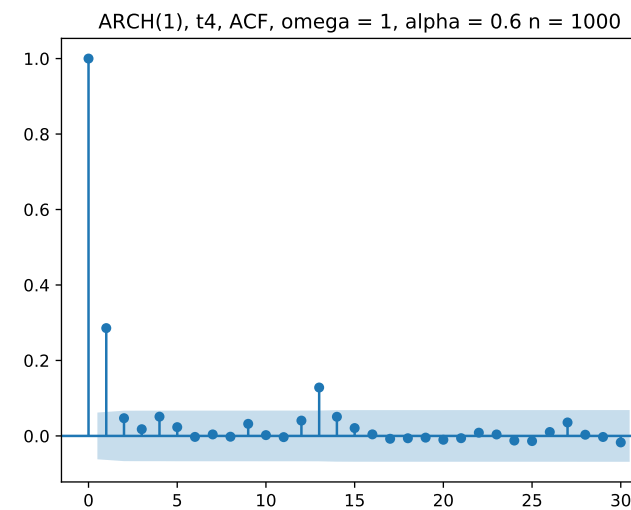
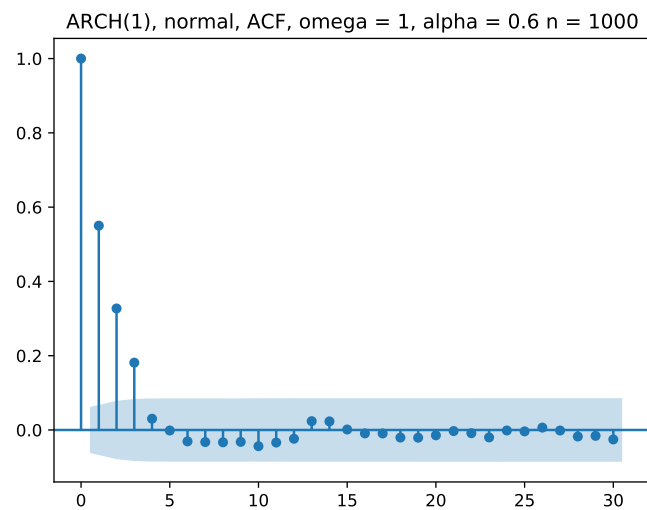
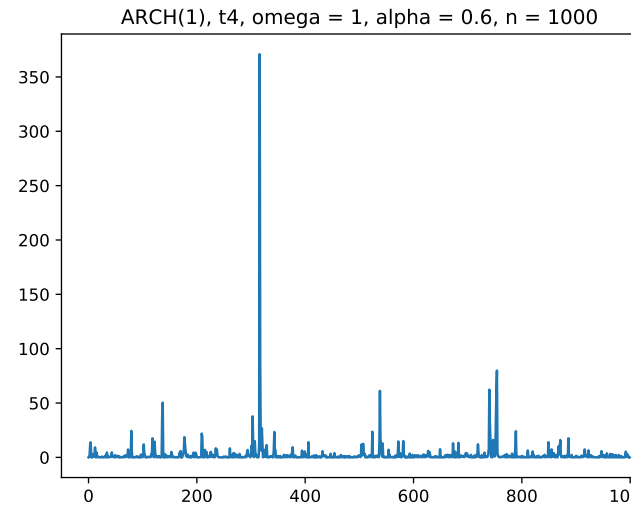
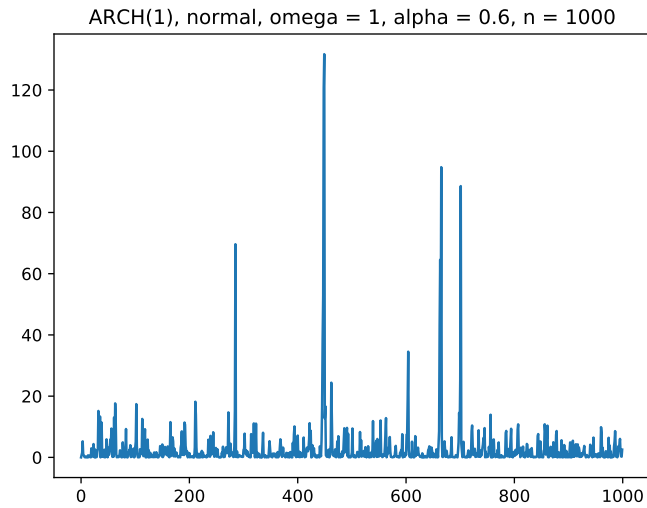
where $\{\epsilon_t\}$ are i.i.d. $N(0,1)$ random variables. Simulate and plot $\{a_t^2, 1 \leq t \leq 1000\}$ assuming $\alpha = .2, .4, .6$. How, if at all, do the plots differ? Do the plots change if the $\text{Normal}(0,1)$ distribution is changed to a $t_4/\sqrt{2}$ distribution? (Note, the t_4 distribution has variance 2).

Solution: We present the plots of $\{a_t^2\}$ for both the normal and the $t_4/\sqrt{2}$ innovations and the program below that also provides the acf and pacf plots. We show just the $\{a_t^2\}$ and the acf plots. Plots for $\alpha = 0$ are included, as there are no ARCH effects for this value. The scale is larger for the t-distribution than for the normal and the number of clear volatility clusters increases as alpha increases.









```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels as sm
```

```
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
```

```
n = 1000
omega = 1
```

```

alpha = 0.6

epn = np.random.randn(n)
ept = np.random.standard_t(4,n)/np.sqrt(2)
signo = np.zeros(n)
sigt = np.zeros(n)
an = np.zeros(n)
at = np.zeros(n)

for i in range(1,n):
    signo[i] = np.sqrt(omega + alpha*an[i-1]**2)
    an[i] = epn[i]*signo[i]
    sigt[i] = np.sqrt(omega + alpha*at[i-1]**2)
    at[i] = ept[i]*sigt[i]

ansq = an*an
atsq = at*at

plt.plot(ansq)
plt.title('ARCH(1), normal, omega = 1, alpha = 0.6, n = 1000')
plt.show()

plot_acf(ansq)
plt.title('ARCH(1), normal, ACF, omega = 1, alpha = 0.6 n = 1000')
plt.show()

plot_pacf(ansq)
plt.title('ARCH(1), normal, PACF, omega = 1, alpha = 0.6, n = 1000')
plt.show()

plt.plot(atsq)
plt.title('ARCH(1), t4, omega = 1, alpha = 0.6, n = 1000')
plt.show()

plot_acf(atsq)
plt.title('ARCH(1), t4, ACF, omega = 1, alpha = 0.6 n = 1000')
plt.show()

plot_pacf(atsq)
plt.title('ARCH(1), t4, PACF, omega = 1, alpha = 0.6, n = 1000')
plt.show()

```

3. (AR(1) + ARCH(1) Process)

Suppose that ϵ_t is an iid $WN(0, 1)$ process, that

$$a_t = \epsilon_t \sqrt{1.5 + 0.30a_{t-1}^2},$$

and that

$$Y_t = 1 + 0.6Y_{t-1} + a_t.$$

- a) Find the mean of Y_t .
- b) Find the variance of Y_t .
- c) Find the autocorrelation function of Y_t .
- d) Find the autocorrelation function of a_t .
- e) Find the autocorrelation function of a_t^2 .

Solutions

- a,b) The Y_t process is an AR(1) process with an ARCH(1) noise process with mean 0 and variance σ_a^2 . Consequently, we may use previously derived formulas, except we first need to determine σ_a^2 . For an ARCH(1) process, $Var(a_t) = \omega/(1.5 - \alpha_1) = 1.5/(1 - .30) = 1.5/.70 = 2.14286$ Consequently

$$E(Y_t) = \phi_0/(1 - \phi_1) = 1/.40 = 2.5.$$

$$Var(Y_t) = \sigma_a^2/(1 - \phi_1^2) = 2.14286/(1 - .6^2) = (1.5/.70)/(1 - .6^2) = 3.3482$$

- c) The autocorrelation function also does not depend upon the specific white noise process.
 $\rho_Y(h) = \phi_1^{|h|}$, so $\rho_Y(h) = .60^{|h|}$
 - d) The ARCH(1) process $\{a_t\}$ is a white noise process. Hence its ACF, $\rho(h) = 1$ for $h = 0$ and 0 otherwise.
 - e) $\{a_t^2\}$ is an AR(1) process whose ϕ_1 parameter is $\alpha = 0.30$. Consequently $\rho_{a^2}(h) = (.3)^{|h|}$.
 process is $\alpha_1^{|h|}$ which is $.30^{|h|}$.
4. (GARCH Modeling)
- Using the S&P 500 index data from problem 1, fit a constant mean μ and GARCH volatility model using the ARCH python package. Note the coefficients, their standard error and the p-values for testing if any of them are 0.

Refer to the documents and examples for the ARCH package at
<https://arch.readthedocs.io/en/latest/univariate/introduction.html>.

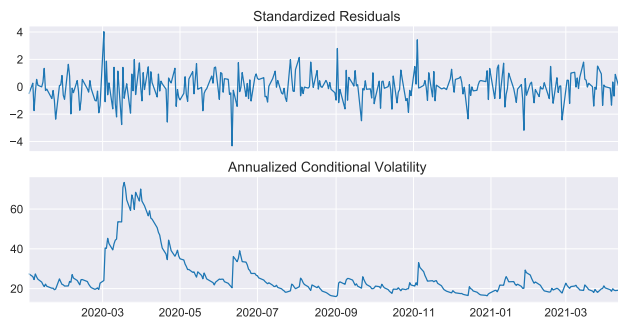
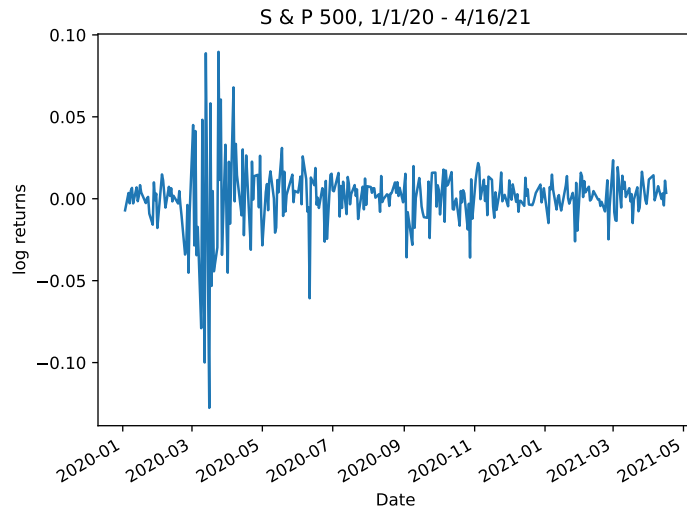
This type of problem is covered in the examples.

Solution Following the example given in class (and code at the bottom), the graph of the log returns is given below followed by the standardized residuals and annualized conditional volatility derived from a GARCH(1,1) model with parameter values

$$\omega = 0.1381(7.424e - 01)$$

$$\alpha = 0.1193(4.282e - 02)$$

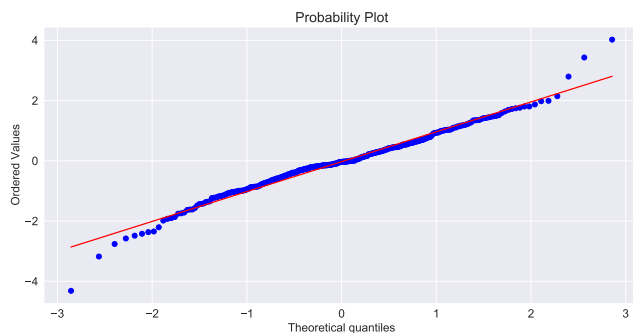
$$\beta = 0.8275(5.793e - 02)$$



We check for remaining ARCH effects using the command

```
smm.stats.diagnostic.het_arch(std_resid)
```

(8.556480730466218, 0.9531576300464633, 0.4873959829114462, 0.9577436943534197)
 The p-values are .953 and .958 for the Engle and the F-test respectively indicating that the ARCH effects have been successfully removed. Finally, we present a probability plot of the standard residuals. Note that there is some evidence of a heavier than normal tail, but not severely so.



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Homework #5

1. GARCH-in-Mean

The text by Hamilton (p667) states “Finance theory suggests that an asset with a higher perceived risk would pay a higher return on average. For example, let Y_t denote the ex-post rate of return on some asset minus the return on a safe alternative asset. Suppose that Y_t is decomposed into a component anticipated by investors at time $t - 1$ (denoted by m_t) and a component that was unanticipated (denoted by a_t):

$$Y_t = m_t + a_t.$$

Then the theory suggests that the mean return (m_t) would be related to the variance of the return σ_t^2 . In general, the ARCH-M (GARCH-M) called ARCH-in-mean (GARCH-in-mean) is given by:

$$Y_t = \beta_0 + \beta_1 X_t + \delta \sigma_t^2 + a_t,$$

$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = 1 + 0.4a_{t-1}^2.$$

Here the $\{\epsilon_t$ are i.i.d. $N(0,1)$ random variables.” The ARCH/GARCH-in-mean model has the additional feature that the conditional volatility appears in the mean model. Assume that $\beta_0 = .06$, $\beta_1 = 0.30$, and $\delta = 0.20$, and the process $\{X_t\}$ is the inflation rate during this time period.

- a) What is $E(Y_t|X_t = 0.1, a_{t-1} = 0.6)$?
- b) What is $Var(Y_t|X_t = 0.1, a_{t-1} = 0.6)$?
- c) Is the conditional distribution of Y_t given X_t and a_{t-1} normal? Why or why not?
- d) Is the marginal distribution of Y_t normal? Why or why not?

Solution

a)

$$\begin{aligned} E(Y_t|X_t, a_{t-1}) &= E(\beta_0 + \beta_1 X_t + \delta \sigma_t^2 + a_t|X_t, a_{t-1}) \\ &= \beta_0 + \beta_1 X_t + \delta(1 + .4a_{t-1}^2) + E(a_t|X_t, a_{t-1}) \\ &= \beta_0 + \beta_1 X_t + \delta(1 + .4a_{t-1}^2) + E(\epsilon_t \sqrt{1 + .4a_{t-1}^2}|X_t, a_{t-1}) \\ &= \beta_0 + \beta_1 X_t + \delta(1 + .4a_{t-1}^2) + E(\epsilon_t)E(\sqrt{1 + .4a_{t-1}^2}|X_t, a_{t-1}) \\ &= \beta_0 + \beta_1 X_t + \delta(1 + .4a_{t-1}^2) + 0 \\ &= 0.06 + 0.30(0.1) + 0.20(1 + .4 * 0.6^2) \\ &= .3188 \end{aligned}$$

b)

$$\begin{aligned} Var(Y_t|X_t, a_{t-1}) &= Var(\beta_0 + \beta_1 X_t + \delta \sigma_t^2 + a_t|X_t, a_{t-1}) \\ &= Var(\delta \sigma_t^2 + a_t|X_t, a_{t-1}) \\ &= Var(a_t|a_{t-1}) \end{aligned}$$

$$\begin{aligned}
&= \text{Var}(\epsilon_t \sigma_t | a_{t-1}) - 0 \\
&= E(\epsilon_t^2 (\omega + .4a_{t-1}^2) | a_{t-1}) \\
&= 1 * (\omega + .4a_{t-1}^2) \\
&= 1 + .4 * .6 * .6 \\
&= 1.144
\end{aligned}$$

- c) Note that conditional on X_t and a_{t-1} , $\beta_0 + \beta_1 X_t + \delta \sigma_t$ is constant, since $\sigma_t = \sqrt{\omega + .4a_{t-1}^2}$. Thus the conditional distribution of Y_t is a constant + $\sqrt{\omega + .4a_{t-1}^2} \epsilon_t$ which has a normal distribution because conditionally on a_{t-1} , σ_t is a constant.
- d) First, the probability distribution of the inflation rate, X_t , is not given. Consequently, one cannot conclude that the distribution of Y_t will be normal. But going further, recall that the kurtosis of a_t is larger than 3, so the noise term in the definition of the Y_t model is not normal, and so unconditionally Y_t will not be normal.
2. Download stock data Merck (ticker = MRK) from 1/1/2011 to 4/23/2021. Run through a complete analysis of time series for the log-returns of this data series.
- Plot the log-returns and analyze the mean model. Check for stationarity using the adjusted Dickey Fuller test. Make any necessary transformations to make the series stationary.
 - Look at the ACF and PACF plots and test for any significant autocorrelation using the Ljung-Box test. Report your results.
 - If there are any significant autocorrelations, find a good ARMA model for these data.
 - Test for any ARCH effects using the Ljung-Box test and the Engle test.
 - If there are any significant ARCH effects, fit a GARCH model to the data.
 - Using the data up to 4/1/21, forecast the volatility going forward to 4/23/2021.
 - Suppose you own 100 shares of MRK on 4/1/2021. What is a 99% confidence interval for the price of MRK on 4/23/2020 based on the models you have developed?

Solution

```

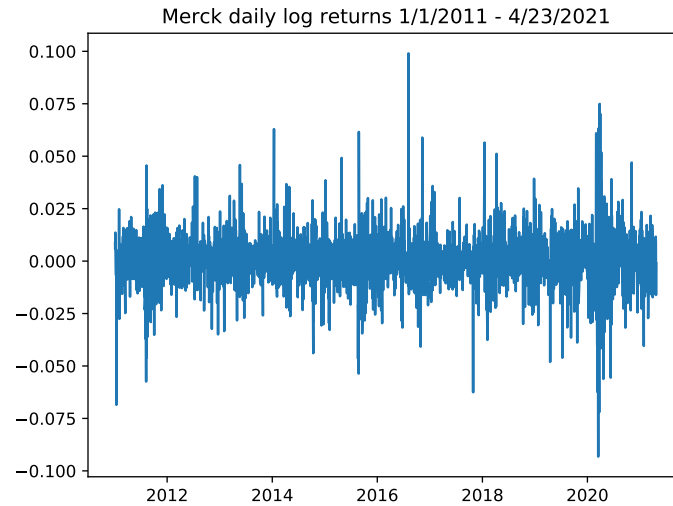
a,b) import pandas as pd
import pandas_datareader as pdr
import numpy as np
import statsmodels.api as smm
import statsmodels.formula.api as smf
import statsmodels.tsa.stattools as smt
import scipy.stats as scs
import matplotlib.pyplot as plt
from statsmodels.stats.diagnostic import het_arch
from arch import arch_model
from scipy.stats import norm, t

start = '2011-01-01'
end = '2021-4-23'

price_data = pdr.get_data_yahoo('mrk', start, end)
price_data_adjclose = price_data['Adj Close']
logret = np.log(price_data_adjclose/price_data_adjclose.shift(1)).dropna()

```

```
smt.adfuller(logret)
```



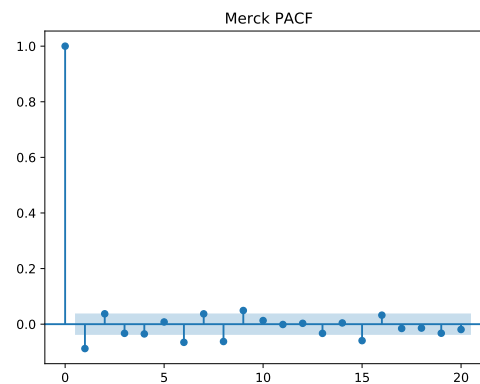
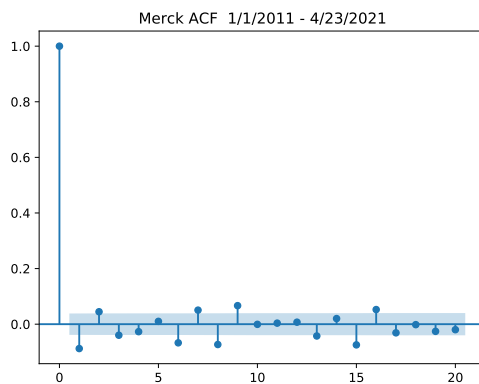
```
smt.adfuller(logret)
```

```
(-13.533680410315649, 2.5922903849253684e-25, 15, 2577, {'1%': -3.4328900953577164, '5%': -2.5758293036286004, '10%': -2.229448912347671})
```

The Dickey Fuller test clearly rejects the null hypothesis (unit root) statistic of -13.53 gives p-value of 0. - thus the series is stationary and no differencing is required.

```
smm.graphics.tsa.plot_acf(logret,lags = 20,alpha = .05)
```

```
smm.graphics.tsa.plot_pacf(logret,lags = 20,alpha = .05)
```



```
smm.stats.diagnostic.acorr_ljungbox(logret,10)
```

```
(array([19.96204106, 25.15249829, 29.24546171, 31.1295321 , 31.39742385, 43.1001994 ,
49.743647 , 63.67265246, 75.24745203, 75.24832642]), array([7.89949054e-06, 3.45306213e-
06, 1.98862024e-06, 2.88066860e-06, 7.81803020e-06, 1.11437349e-07, 1.62190664e-08,
8.82678962e-11, 1.41212742e-12, 4.25660761e-12]))
```

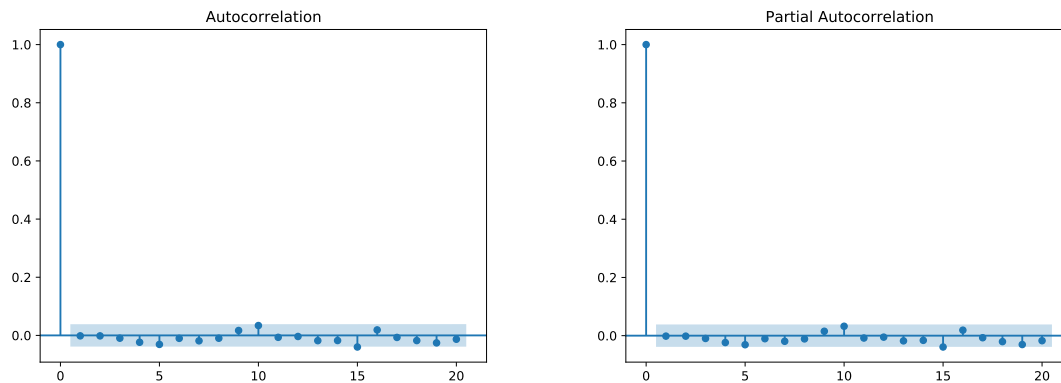
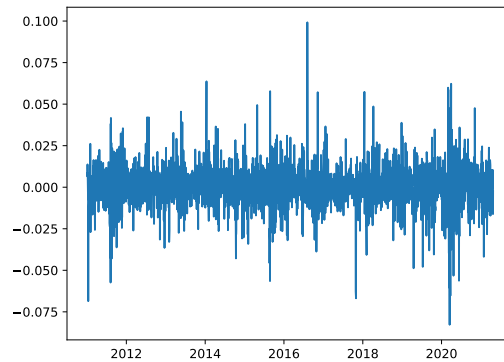
Clearly there are significant autocorrelations with very tiny p-values.

c) Using the ARMA order select command:

```
res_aic = smm.tsa.arma_order_select_ic(logret,ic='aic', trend='nc')
```

```
print(res_aic.aic_min_order)
res_bic = smm.tsa.arma_order_select_ic(logret,ic='bic', trend='nc')
print(res_bic.bic_min_order)
```

Both the AIC and the BIC as select an ARMA(4,2) model.



This shows that the temporal effects have been modeled. However, we need to check for ARCH effects.

d) Both the Ljung-Box test and the Engle test confirm the existence of ARCH effect.

First, the Ljung-Box test:

```
smm.stats.diagnostic.acorr_ljungbox(model.resid**2,lags = 10)
```

```
(array([ 68.06280581, 138.85000818, 180.98320213, 195.94145383,
        269.25361566, 276.10938136, 300.03560939, 359.01534258,
        366.77811932, 433.52745577]), array([1.58369963e-16, 7.06486418e-31, 5.4092400e-35,
        4.04772673e-56, 1.06898937e-56, 5.94463492e-61, 1.07692330e-72,
        1.65816714e-73, 6.79978709e-87]))
```

Next the Engle test for ARCH Effects:

```
print(het_arch(model_aic.resid**2, 10))
(62.38167053966375, 1.2803379499676652e-09, 6.365329282611634, 9.86718913699239e-10)
```

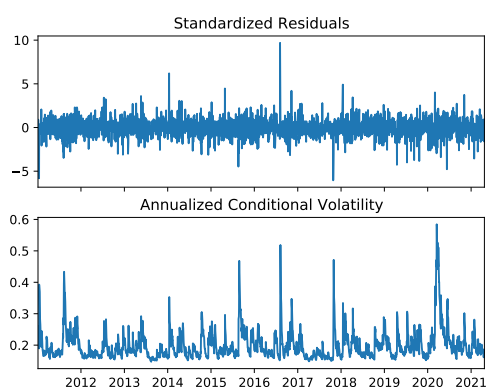
Both given very strong evidence for the presence of ARCH effects

```
e) am = arch_model(model.resid)
res.summary()
```

Constant Mean - GARCH Model Results					
=====					
Dep. Variable:	None	R-squared:	0.000		
Mean Model:	Constant Mean	Adj. R-squared:	0.000		
Vol Model:	GARCH	Log-Likelihood:	7758.60		
Distribution:	Normal	AIC:	-15509.2		
Method:	Maximum Likelihood	BIC:	-15485.8		
		No. Observations:	2593		
Date:	Sat, May 01 2021	Df Residuals:	2592		
Time:	23:28:14	Df Model:	1		
Mean Model					
=====					
	coef	std err	t	P> t	95.0% Conf. Int.

mu	7.9712e-04	2.447e-04	3.258	1.122e-03	[3.176e-04,1.277e-03]
Volatility Model					
=====					
	coef	std err	t	P> t	95.0% Conf. Int.

omega	1.6482e-05	2.853e-12	5.777e+06	0.000	[1.648e-05,1.648e-05]
alpha[1]	0.1000	2.787e-02	3.589	3.326e-04	[4.538e-02, 0.155]
beta[1]	0.8000	2.760e-02	28.981	1.155e-184	[0.746, 0.854]



To check if the ARCH effects have been accounted for, we run the Engle test
`statsmodels.stats.diagnostic.het_arch(std.resid)`

(7.459031674551876, 0.9999633193158198, 0.2641502767269545, 0.9999651407360753)

The p-values for the Engle test and the F test are essentially 1.0 giving strong assurance that the ARCH effects have been addressed by the GARCH(1,1) model with ω nearly 0, $\alpha = 0.1$ and $\beta = 0.8$. The persistence parameter $\lambda = \alpha + \beta = .9$

One can also run the Ljung-Box test on the squared standardized residuals. The results also give very strong evidence that the standardized squared residuals are white noise indicating that the ARCH effects have been fully modeled.


```
statsmodels.stats.diagnostic.acorr_ljungbox(std_resid**2, lags = 15)
```

```
(array([ 0.40994019,  0.46671663,  0.54511048,  1.99915097,  2.12803247,
        2.69836204,  2.9423119 ,  3.03226827,  3.75596718,  3.79287652,
        3.92384888,  3.94044916,  3.95461641,  4.18713149,  4.22266757,
        4.47974413,  4.53199072,  5.10715822,  5.21170996,  5.29994968,
        5.36275621,  5.40454748,  6.73665439,  6.84335645,  6.94854879,
        7.83200324,  8.17951689,  8.1935596 , 10.12928198, 10.12929825]), array([0.52
        0.84564073, 0.89028705, 0.93231623, 0.92672764, 0.9562073 ,
        0.97209185, 0.98448713, 0.99166462, 0.99423118, 0.99691202,
        0.99779409, 0.99881503, 0.99868321, 0.99923142, 0.99956528,
        0.99976552, 0.99987908, 0.99960314, 0.99976362, 0.99986044,
        0.99977584, 0.99981874, 0.99990236, 0.99954509, 0.99974142]))
```

- f) This part of the problem involves using data up to 4/1/21 to forecast volatility from 4/2/21 to 4/23/21. We need to be careful to use only the data up to 4/1/21 and refit the models using the slightly smaller dataset to avoid “look ahead bias.”

We execute the same commands as before on the smaller data set (not shown here): (getting variables logret1, model1, am1 and res1). The mean model is still best fit by an ARMA(4,2) model (AR IMA(4,0,2)). The GARCH volatility is also fit by a GARCH(1,1) model with nearly identical parameter values.

	Volatility
Date	
2021-04-02	0.011532
2021-04-05	0.011680
2021-04-06	0.011812
2021-04-07	0.011929
2021-04-08	0.012034
2021-04-09	0.012127
2021-04-12	0.012211
2021-04-13	0.012285
2021-04-14	0.012352
2021-04-15	0.012412
2021-04-16	0.012466
2021-04-19	0.012514
2021-04-20	0.012557
2021-43-21	0.012595
2021-04-22	0.012630

- g) Recall that problem 2g was cancelled.

Financial Time Series

46-929

Solutions #6

1. Download from canvas the file fredgraph.csv. This file contains data from the St. Louis Federal Reserve giving the U.S. Treasury monthly average constant maturity rates for 1 year, 5 years and 10 years dating back to 1953. Plot these variables, perform a Dickey Fuller test to check for stationarity. If needed, difference the data to transform it to $\mathcal{I}(1)$. Perform cointegration tests to check if any pairs of these data are cointegrated, following what was done in class or equivalently follow the example give in

https://bashtage.github.io/arch/doc/unitroot/unitroot_cointegration_examples.html

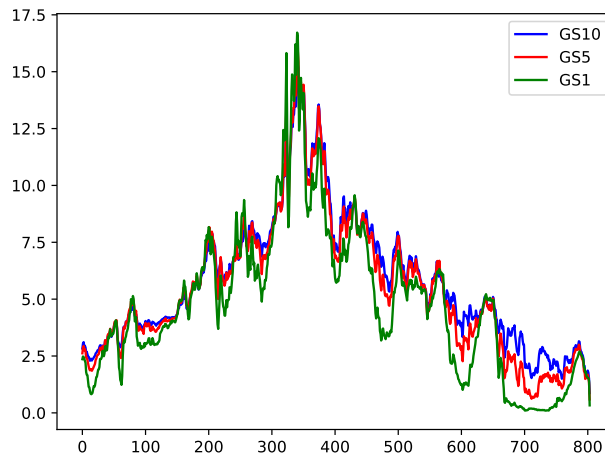
Solution

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from arch.unitroot import engle_granger
from arch.unitroot.cointegration import phillips_ouliaris
from statsmodels.tsa.vector_ar.vecm import coint_johansen

import numpy as np
import statsmodels as smm
import statsmodels.api as sm
import statsmodels.formula.api as smf
import statsmodels.tsa as smt
import scipy.stats as scs
import matplotlib.pyplot as plt

data = pd.read_csv('fredgraph.csv')
data.head()
```



```

smm.tsa.stattools.adfuller(data['GS10'])
(-0.867, 0.80, 21, 782, '1%': -3.44, '5%': -2.87, '10%': -2.59, 38.87)
smm.tsa.stattools.adfuller(data['GS5'])
(-1.06, 0.73, 21, 782, '1%': -3.44, '5%': -2.87, '10%': -2.57, 235.69)
smm.tsa.stattools.adfuller(data['GS1'])
(-1.77, 0.39, 20, 783, '1%': -3.44, '5%': -2.87, '10%': -2.57, 556.53)

```

This shows that the H_0 in the Dickey Fuller test is NOT rejected, hence we cannot conclude that any of the three series are stationary. Hence we difference each series one time.

```

diff10 = np.diff(data['GS10'])
diff5 = np.diff(data['GS5'])
diff1 = np.diff(data['GS1'])
smm.tsa.stattools.adfuller(diff10)
(-6.89, 1.37e-09)
smm.tsa.stattools.adfuller(diff5)
(-6.81, 2.08e-09)
smm.tsa.stattools.adfuller(diff1)
(-6.78, 2.47e-09)

```

In all three cases, the null hypothesis of a unit root is overwhelmingly rejected. So each of the differenced series is stationary, and the series themselves are all $\mathcal{I}(1)$.

```

eg_test = engle_granger(data['GS10'], data['GS5'], trend="n")
eg_test
Engle-Granger Cointegration Test
Statistic: -3.088906009822177
P-value: 0.021326184167723132
Null: No Cointegration, Alternative: Cointegration

```

```

eg_test = engle_granger(data['GS10'], data['GS1'], trend="n")
eg_test
Engle-Granger Cointegration Test
Statistic: -3.4015501707752556

```

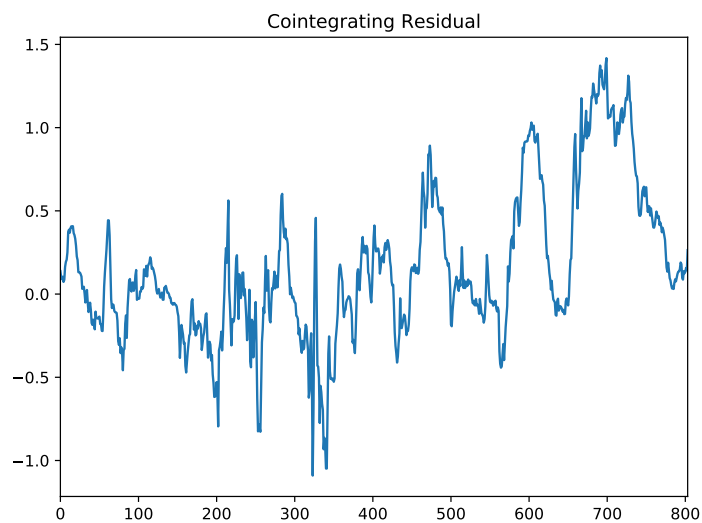
P-value: 0.008451634515862455
Null: No Cointegration, Alternative: Cointegration

```
eg_test = engle_granger(data['GS5'], data['GS1'], trend="n")  
eg_test  
Engle-Granger Cointegration Test  
Statistic: -3.8647927995305857  
P-value: 0.0017883424447213106  
Null: No Cointegration, Alternative: Cointegration
```

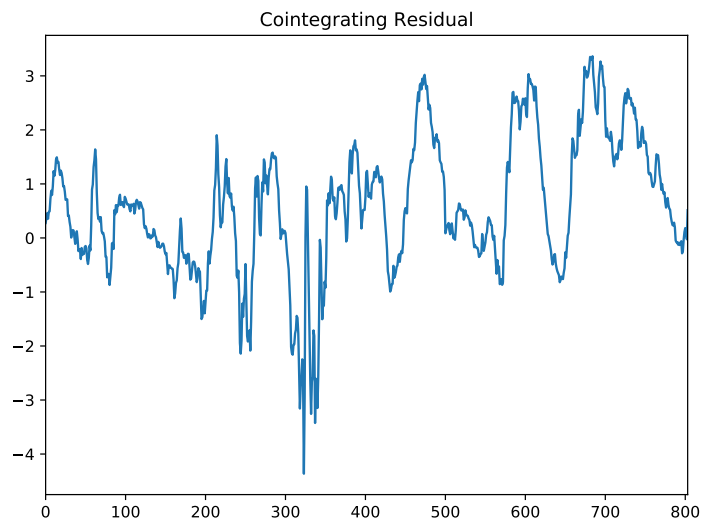
```
fig = eg_test.plot()  
eg_test.cointegrating_vector  
GS10 1.000000  
GS5 -1.027517
```

```
fig = eg_test.plot()  
plt.show()  
eg_test.cointegrating_vector  
GS10 1.000000  
GS1 -1.081807
```

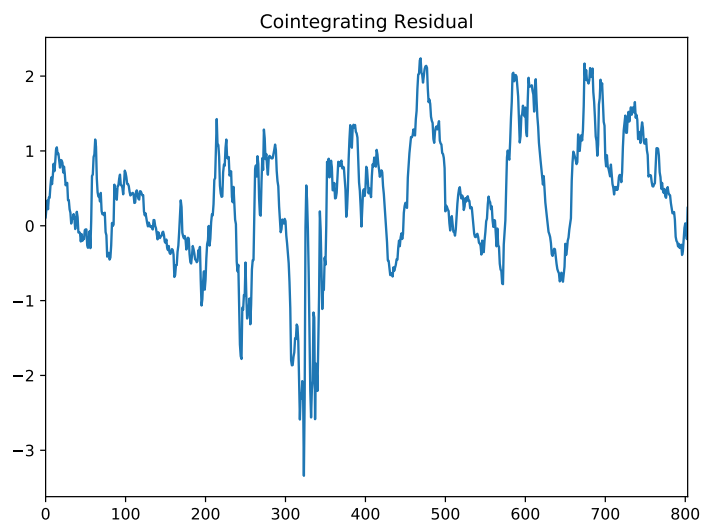
```
fig = eg_test.plot()  
plt.show()  
eg_test.cointegrating_vector  
GS5 1.000000  
GS1 -1.062635
```



Cointegration Residuals: GS10 and GS5



Cointegration Residuals: GS10 and GS1



Cointegration Residuals: GS5 and GS1

These Engle-Granger test show that all pairs of series are cointegrated, it is nearly the case that differencing any of the three pairs gives a stationary process as the cointegrating vectors are all approximately (1,-1). The GS10 and GS5 pair has a p-value of .02, hence we can reject the lack of cointegration. The other two pairs have smaller p-values.

```
po_zt_test = phillips_ouliaris(data['GS10'], data['GS5'], trend="c", test_type="Zt")
po_zt_test.summary()
;class 'statsmodels.iolib.summary.Summary';
```

Phillips-Ouliaris zt Cointegration Test

```
=====
Test Statistic -3.398
P-value 0.043
```

Kernel Bartlett
Bandwidth 12.829

Trend: Constant Critical Values: -3.05 (10%) Null Hypothesis: No Cointegration Alternative Hypothesis: Cointegration Distribution Order: 3

```
po_zt_test = phillips_ouliaris(data['GS10'], data['GS1'], trend="c", test_type="Zt")
po
zt_test.summary()
class 'statsmodels.iolib.summary.Summary'
```

Phillips-Ouliaris zt Cointegration Test

```
=====
Test Statistic      -3.692
P-value              0.019
Kernel               Bartlett
Bandwidth            11.545
=====
```

Trend: Constant
Critical Values: -3.05 (10%), -3.35 (5%), -3.92 (1%)
Null Hypothesis: No Cointegration
Alternative Hypothesis: Cointegration
Distribution Order: 3

```
po_zt_test = phillips_ouliaris(data['GS5'], data['GS1'], trend="c", test_type="Zt")
po_zt_test.summary()
class 'statsmodels.iolib.summary.Summary'
```

Phillips-Ouliaris zt Cointegration Test

```
=====
Test Statistic      -4.244
P-value              0.003
Kernel               Bartlett
Bandwidth            16.167
=====
```

Trend: Constant
Critical Values: -3.05 (10%), -3.35 (5%), -3.92 (1%)
Null Hypothesis: No Cointegration
Alternative Hypothesis: Cointegration
Distribution Order: 3

The Phillips-Ouliaris test also show cointegration of all the pairs, although the p-value for the GS10 and GS5 pair has a p-value of .04 and so is barely significant. The others are more clearly significant, thus the Phillips-Ouliaris test agrees with the Engle Granger test. The Johansen test was also run for both the trace test and the maximum eigenvalue test.

```
johansen_rst = coint_johansen(data, det_order = 0, k_ar_diff = 0)
johansen_rst.lr1
trace statistic: array([53.44485201, 18.08751136, 3.65095396])
johansen_rst.lr2
maximum eigenvalue statistic: array([35.35734065, 14.43655739, 3.65095396])
```

```
johansen_rst.cvt
array(((27.0669, 29.7961, 35.4628),
(13.4294, 15.4943, 19.9349),
( 2.7055, 3.8415, 6.6349)))
johansen_rst.cvm
array(((18.8928, 21.1314, 25.865 ),
(12.2971, 14.2639, 18.52 ),
( 2.7055, 3.8415, 6.6349)))
```

According to the trace statistic there is one cointegrating vector at the 1% level and two at the 5% level. The same is true using the maximum eigenvalue statistic.

2. Consider a univariate AR(p) process

$$X_t - \mu = \phi_1(X_{t-1} - \mu) + \cdots \phi_p(X_{t-p} - \mu) + \epsilon_t.$$

Write this in state space form, identifying all of the components of that form taking $\mathbf{X}_t = (X_t - \mu, \dots, X_{t+1-p} - \mu)$ (p-dimensional) and X_t (1-dimensional) as the variable in the observation equation.

Solution

We can put the AR(1) model into state space form for the processes

$$\mathbf{X}_t = \mathbf{F}_t \mathbf{X}_{t-1} + \mathbf{V}_t,$$

$$\mathbf{Y}_t = \mathbf{G}_t \mathbf{X}_t + \mathbf{W}_t,$$

with $\mathbf{Y}_t = (1, \dots, 0) \mathbf{X}_t$, $\mathbf{W}_t = \mathbf{0}$, $\mathbf{V}_t = (\epsilon_t + \mu(1 - \sum_{i=1}^p \phi_i), 0, \dots, 0)^T$ $\mathbf{X}_t = (X_t, \dots, X_{t+1-p})^T$ and

$$F = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \cdots & \phi_p \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & & 0 \\ 0 & & & 1 & 0 \end{bmatrix}.$$

3. Consider the local trend model

$$M_{t+1} = M_t + V_t,$$

$$Y_t = M_t + W_t,$$

where $\{V_t\}$ is $\text{WN}(0, \sigma_V^2)$ and $\{W_t\}$ is $\text{WN}(0, \sigma_W^2)$

- a) Show that Y_t can be written as an ARIMA(0,1,1) model

$$Y_t = Y_{t-1} + Z_t + \theta Z_{t-1},$$

for an appropriately chosen noise sequence Z_t , where $\{Z_t\}$ is $\text{WN}(0, \sigma^2)$.

- b) Determine possible solutions for θ and σ in terms of σ_V^2 and σ_W^2 .
- c) Let $\sigma_W^2 = 8$ and $\sigma_V^2 = 20$, simulate a series of Y_t according to this local trend model for the possible solutions found in part b) with these specific values for σ_W^2 and σ_V^2 .
- d) Perform a time series analysis on the simulated Y_t series. What do you conclude?

Solution

Note, Problem 2 is about putting a standard time series model into state space form. This problem is taking a model in state space form and putting it into traditional form. There are two possible solutions.

- a) To show that Y_t is an ARIMA(0,1,1) series, look at the autocorrelation of the first difference series. We first determine the autocovariance function of the $\nabla Y_t = Y_t - Y_{t-1}$ process. $Y_t - Y_{t-1} = M_t - M_{t-1} + W_t - W_{t-1} = V_{t-1} + W_t - W_{t-1}$. $\gamma(0) = \text{Var}(V_{t-1} + W_t - W_{t-1}) = \sigma_V^2 + 2\sigma_W^2$. Similarly, $\gamma(1) = \text{Cov}(V_{t-1} + W_t - W_{t-1}, V_{t-2} + W_{t-1} - W_{t-2}) = -\sigma_W^2$. Note that $\gamma(h) = 0$ for $h \geq 2$. Now, for a MA(1) process where $Y_t - Y_{t-1} = Z_t + \theta Z_{t-1}$, $\gamma(0) = \text{Var}(Z_t + \theta Z_{t-1}) = \sigma^2(1 + \theta^2)$. $\gamma(1) = \text{Cov}(Z_t + \theta Z_{t-1}, Z_{t-1} + \theta Z_{t-2}) = \theta\sigma^2$. Furthermore, $\gamma(h) = 0$ for $h \geq 2$.
- b) We have two simultaneous equations, $\theta\sigma^2 = -\sigma_W^2$ and $(\theta^2 + 1)\sigma^2 = \sigma_V^2 + \sigma_W^2$. Substituting $\sigma^2 = -\sigma_W^2/\theta$ we find that θ must satisfy a quadratic equation $(1 + \theta^2)/\theta = -(\sigma_V^2 + 2\sigma_W^2)/\sigma_W^2$. This is a simple quadratic equation that can be explicitly solved for (θ, σ^2) in terms of (σ_V^2, σ_W^2) . There will be two real solutions.
- c) If $\sigma_W^2 = 8$ and $\sigma_V^2 = 20$, there are two solutions $(\sigma^2 = 1.8754, \theta = -4.2656)$ and $(\sigma^2 = 34.1245, \theta = -.2344)$.
- d) Although the two parameter sets are quite different, the time series displays (time series plot, ACF and PACF) are essentially indistinguishable looking just at the second order properties.

